
The Acrocentric Short Arms

Sequence, function, assembly, alignment and variant analysis of the last 66 Mb of the human genome — a consolidated review, with an agenda

The five acrocentric short arms comprise 66.07 Mb of T2T-CHM13v2.0 and 2.1% of the human genome. GRCh38 represents 83% of that territory as N; only 4.3% of it is short-read accessible, against 86.9% of the autosomal genome; between 22% and 61% of each arm has a $\geq 99\%$ -identical copy on a different acrocentric chromosome; and the de novo single-nucleotide mutation rate within it is roughly tenfold the autosomal euchromatic rate. This report consolidates what is known about these regions, sets out how to assemble, align and analyse them from single nucleotides to whole-arm rearrangements, and proposes specific solutions to the coordinate, assignability, representation and benchmarking problems that currently prevent them from being studied at scale.

Prepared 18 September 2026.

All quantitative statements about reference content were computed for this report from the CenSat v2.1 satellite annotation, the SEDEF segmental-duplication call set, the short-read accessibility mask and the minimum-unique- k -mer-length tracks distributed for T2T-CHM13v2.0. Methods in section 25; data resources in section 27.

11 figures, 8 tables, 109 references.

Contents

Executive summary	4
Twelve findings that should change how these regions are analysed	4

Part I • The object 8

1. Nomenclature, and why the vocabulary matters	8
2. A quantitative anatomy	8
3. The compartments, one by one	10
3.1 Telomere and the distal p13 subtelomeric region	10
3.2 The distal junction (DJ)	10
3.3 The rDNA array	11
3.4 The proximal junction (PJ)	12
3.5 β -satellite (Sau3A / BSR)	12
3.6 HSat1A and HSat1B	12
3.7 HSat2 and HSat3	12
3.8 SST1 (NBL2) — the recombinogenic macrosatellite	13
3.9 α -satellite: monomeric, divergent HOR, and active HOR	13
3.10 Segmental duplications and the acrocentric "community"	13
4. Function	14
4.1 Ribosome biogenesis	14
4.2 Nucleolar organisation and three-dimensional genome architecture	14
4.3 Centromere function	15
4.4 Satellite transcription and heterochromatin	15
4.5 Meiotic behaviour	15
5. Variation, population genetics and evolution	15
5.1 What varies, and by how much	16
5.2 Hybrid arms, PHRs and ectopic exchange	16
5.3 De novo mutation	16
5.4 Where the elevated mutation rate comes from	16
5.5 Is the order of the blocks the same in everyone?	18
5.6 Evolution	20
6. Disease	21
6.1 Robertsonian translocations	21
6.2 Aneuploidy, uniparental disomy and imprinting	21
6.3 rDNA in cancer and ageing	21
6.4 What is plausibly being missed in rare-disease cohorts	21

Part II • Working with them 23

7. References and coordinate systems	23
--	----

7.1 The choice of reference is the single largest determinant of what you can see	23
7.2 Practical guidance on reference choice	24
7.3 Liftover and coordinate translation	24
7.4 Masks you should be carrying	24
8. Data generation: what to buy for what question	24
8.1 The read-length ladder, read as a purchasing decision	25
8.2 Minimum viable designs	26
9. Assembly	26
9.1 What current assemblers actually achieve	26
9.2 Assembler choice	26
9.3 Identifying and anchoring short-arm contigs	27
10. Validation and quality control	27
11. Alignment, and the assignability problem	28
11.1 Mappability is not assignability	28
11.2 Strategies that work	28
12. Annotation	28
Part III • Variants, from single nucleotides to whole arms	30
13. A three-regime framework, and why VCF breaks	30
13.1 The framework	30
13.2 Why the VCF model fails in Regimes II and III	30
13.3 The decision matrix	31
14. SNVs and indels, including de novo calling in trios	32
14.1 What is callable, and how to define it	32
14.2 De novo SNV calling: the procedure that works	32
14.3 What fails, and the specific false-positive classes	32
14.4 Recalibrating expectations	32
15. Tandem-repeat and array-level variation	33
16. rDNA copy number and sequence	33
17. Large structural variants, NAHR and whole-arm events	34
18. Somatic and cancer applications	34
19. Short-read cohorts: what is actually possible	34
Part IV • Challenges, proposals and an agenda	36
20. Twelve unresolved challenges	36
21. Connections we think are real but unproven	37
22. Seven concrete proposals, and one of them built	38
22.1 Anchor-relative coordinates for the short arms	38
22.2 An acrocentric pangenome module with an explicit ambiguity model	38

22.3 An assignability score to replace MAPQ	39
22.4 An array-level variant representation	39
22.5 Per-chromosome rDNA copy number by IGS-barcode deconvolution	39
22.6 A sequencing-based Robertsonian carrier test, and an acrocentric liability score	40
22.7 An acrocentric benchmark	40
22.8 Implemented: a learned slot grammar, and the placement of unlabelled sequence	40
23. Cookbook	44
23.1 If you have short reads only	44
23.2 If you have HiFi only (~30×)	44
23.3 If you have HiFi + UL-ONT + Hi-C	45
23.4 If you have a trio or pedigree	45
23.5 If you have a Robertsonian or other rearrangement carrier	45
24. Reporting standards for acrocentric analyses	45
25. Methods for the analyses in this report	45
26. Glossary	47
27. Resources	48
28. A closing assessment	48
29. References	49

Executive summary

For thirty years the short arms of the five human acrocentric chromosomes — 13p, 14p, 15p, 21p and 22p, collectively the SAACs — were a blank in the reference genome and a blank in nearly every analysis built on it. They are no longer a blank. The T2T-CHM13 assembly supplied sequence¹, the Human Pangenome and HGSC efforts supplied population-scale context^{2–3}, pangenome-graph analysis supplied the first map of exchange between non-homologous arms⁴, and a four-generation pedigree has now supplied the first direct measurements of mutation and recombination inside them⁵. What has *not* arrived is a settled analytical practice. Most variant-calling pipelines, most callable-region definitions, and essentially every clinical filter still treat these 66 Mb as if they did not exist.

This report consolidates what is known and proposes how to work on it. Sections 1–6 describe the object: nomenclature, quantitative anatomy, each sequence compartment, biological function, population genetics and disease. Sections 7–12 describe method: references and coordinates, data generation, assembly, validation, alignment, annotation. Sections 13–19 treat variant analysis from single nucleotides to whole-arm rearrangements. Sections 20–24 set out the challenges we consider unresolved, together with specific, testable proposals — several of which are, to our knowledge, new.

All quantitative statements about reference content in this report were computed for it, from the CenSat v2.1 satellite annotation, the SEDEF segmental-duplication calls, the short-read accessibility mask and the minimum-unique-*k*-mer-length tracks distributed for T2T-CHM13v2.0^{6–9}. Methods are given in section 25.

Twelve findings that should change how these regions are analysed

1. **The short arms total 66.07 Mb of sequence** — 2.1% of the genome — measured from the telomere to the start of the active α -satellite higher-order-repeat array on each chromosome. GRCh38 represents that territory with 14.05 Mb of sequence and 66.85 Mb of N: **83% of the cytogenetic short arms is literally absent** from the reference most cohorts are still aligned to.
2. **They are built from one shared parts list, arranged differently on each chromosome.** HSat3 accounts for 15.54 Mb, non-satellite sequence 15.89 Mb, rDNA 9.93 Mb, HSat1A 8.79 Mb, β -satellite 6.12 Mb and α -satellite 5.54 Mb across the

five arms — but 13p is 42% HSat1A while 15p is 52% HSat3 and 22p is 21% β -satellite. There is no "typical" acrocentric short arm.

3. **Being non-satellite does not make sequence tractable.** The 15.89 Mb of non-satellite short-arm DNA is only 4.3% short-read accessible — statistically indistinguishable from HSat3 at 5.5%. The barrier is not tandem repetition; it is duplication *between* arms.
4. **Between 22% and 61% of each short arm has a $\geq 99\%$ -identical copy on a different acrocentric chromosome,** counting only segmental duplications that are not satellite-dominated. 21p leads at 60.5%, 13p at 51.9%. Paralogy between 13p–21p and 14p–22p is the strongest, and the two pairs also share their active centromeric HOR families (S2C13/21H1L and S2C14/22H1L). The cytogeneticists' pairing preferences are written in the sequence.
5. **Only 4.3% of the short arms is short-read accessible, against 86.9% of the autosomal genome.** No compartment exceeds 18%. For short-read cohorts this is not a coverage problem to be solved with more reads; it is a structural ceiling.
6. **Read length buys back most of the arm, then abruptly stops buying.** The fraction of positions with a genome-unique anchor rises from 15.6–45.4% at 150 bp to 77.5–93.6% at 10 kb — and then plateaus. Going from 10 kb to 100 kb adds 1.5–3.4 percentage points. On 13p and 21p, $\sim 19\%$ and $\sim 18\%$ of positions have no unique anchor shorter than 200 kb.
7. **The residual hard core is the rDNA array, and almost nothing else.** Excluding rDNA, every satellite compartment is $>97\%$ anchorable at 10 kb. The rDNA arrays — 9.93 Mb, 221 modelled units in CHM13 — are 0.0% short-read accessible and only 16% anchorable at 100 kb. After long reads, the acrocentric problem is largely an rDNA problem plus an arm-assignment problem, not a per-base mappability problem.
8. **Reference uniqueness is not the same as correct placement.** Mappability computed on CHM13 describes CHM13. Because satellite arrays vary in length by megabases between haplotypes — a single chr15 HSat1A array reaches 1.64 Mb, six times the CHM13 length⁵ — a read can map uniquely and still map to the wrong place. We distinguish *mappability* from *assignability* throughout, and argue that assignability is the quantity that actually limits acrocentric analysis.
9. **Mutation behaves differently here.** The de novo SNV rate on the short arms is 1.33×10^{-7} per bp per generation, roughly tenfold the autosomal euchromatic rate and comparable to Yq12 heterochromatin, with depleted CpG>TpG and excess C>G and A>C substitutions; allelic meiotic crossover is absent from the p-arm while the proximal q-arm is enriched 2.3-fold⁵. Any DNM study that excludes the SAACs is not merely losing 2% of the genome; it is excluding the 2% with the highest per-base yield.
10. **The variant-representation model itself fails here.** For the $\sim 5\text{--}21\%$ of each arm in which no per-base coordinate is portable between individuals, a VCF record is not a description of an allele — it is an artefact of an arbitrary reference path. Section 22 proposes an array-level representation and a benchmark to evaluate it.
11. **The block order is conserved; only the block sizes vary.** Across 944 haplotype-resolved arms — 789 from 94 unrelated HPRC release 2 individuals and 155 from one pedigree — the landmark backbone — ACRO, distal junction, rDNA, proximal junction, centromere — has an order consistency of 0.969, and the modal distal satellite order (β -satellite, HSat3, HSat1A, HSat3) is the same on all five chromosomes. β -satellite always takes the distal-most satellite slot and HSat1A the slot abutting the centromere, in 99–100% of informative arms in both datasets; the least constrained relationship, β -satellite versus HSat1B in the mid-proximal region, sits at 0.64. What varies is size — up to 9.5-fold for HSat3 — not order. Sections 5.5 and 22.8.
12. **The tenfold mutation-rate elevation has two separable components.** Normalising the published de novo counts by callable bases per compartment gives 2.03×10^{-7} in satellite sequence against 0.76×10^{-7} in non-satellite short-arm sequence — a rate ratio of 2.66 (95% CI 1.76–4.02, $P = 2 \times 10^{-6}$). Non-satellite sequence inside the arm is still elevated 5.6-fold over autosomal euchromatin, so a position-linked effect supplies a floor across the whole arm and satellite composition roughly triples it again. Section 5.4.

How to use this report

Readers who need to make an immediate decision about a pipeline should read section 13 (the three-regime framework), section 19 (short-read cohorts), section 23 (cookbook) and Table 6. Readers who need to justify those decisions to a reviewer or a committee should read Part I. Readers looking for research problems should read sections 20–22, where we have tried to separate what is established, what is inferable, and what is speculation — flagged as such.

Part I · The object

What the acrocentric short arms are made of, what they do, and how they vary

1. Nomenclature, and why the vocabulary matters

Human chromosomes 13, 14, 15, 21 and 22 are acrocentric: the centromere sits near one end, leaving a short arm of roughly 10–17 Mb and a long arm carrying essentially all of the protein-coding content. The Y chromosome is sometimes grouped with them, and shares several relevant properties — a large heterochromatic block, satellite families in common, an elevated mutation rate — but it carries no rDNA and is not an acrocentric in the classical sense. Chromosome 15 and 22 also carry α -satellite higher-order-repeat islands within the short arm proper, so "short arm" and "non-centromeric" are not synonyms.

The literature uses several overlapping coordinate vocabularies, and confusion between them is a recurring source of error.

Cytogenetic bands. The short arm resolves into p11.1, p11.2, p12 and p13 from the centromere outward. p12 is, to a first approximation, the rDNA array; p13 is the distal (telomere-proximal) satellite region; p11 is the proximal satellite region abutting the centromere. This vocabulary remains useful because it is what karyotype reports and much of the older literature speak, and because the bands correspond reasonably well to the molecular compartments.

Distal and proximal. Because the rDNA array is the one landmark present on all five arms, the field partitions each arm relative to it: *distal sequence* lies between the telomere and the rDNA, *proximal sequence* between the rDNA and the centromere^{10–11}. The two flanking segmental duplications are named accordingly — the **distal junction (DJ)** and the **proximal junction (PJ)**. Note the polarity convention: rDNA transcription runs *centromere-ward*, so the DJ is downstream of the transcription unit and the PJ upstream of it¹⁰.

Nucleolar organizer region (NOR). The cytological entity — the rDNA array plus its junctions — that nucleates a nucleolus. "NOR" is a functional term; "rDNA array" is a sequence term; they are not interchangeable, since a NOR can be present and transcriptionally silent.

Assembly-era terms. Two are now standard and worth adopting. A **pq-scatig** is a scaffolded contig that spans the short arm, the centromere and at least 1 Mb of uniquely-alignable long arm — the unit of evidence in short-arm assembly work, because the long arm is what allows a short-arm sequence to be assigned to a chromosome at all^{4–5}. A **pseudo-homologous region (PHR)** is a segment shared between non-homologous acrocentric arms at high identity, showing positional-homology patterns consistent with ectopic exchange; the PHRs identified to date centre on the SST1 macrosatellite⁴.

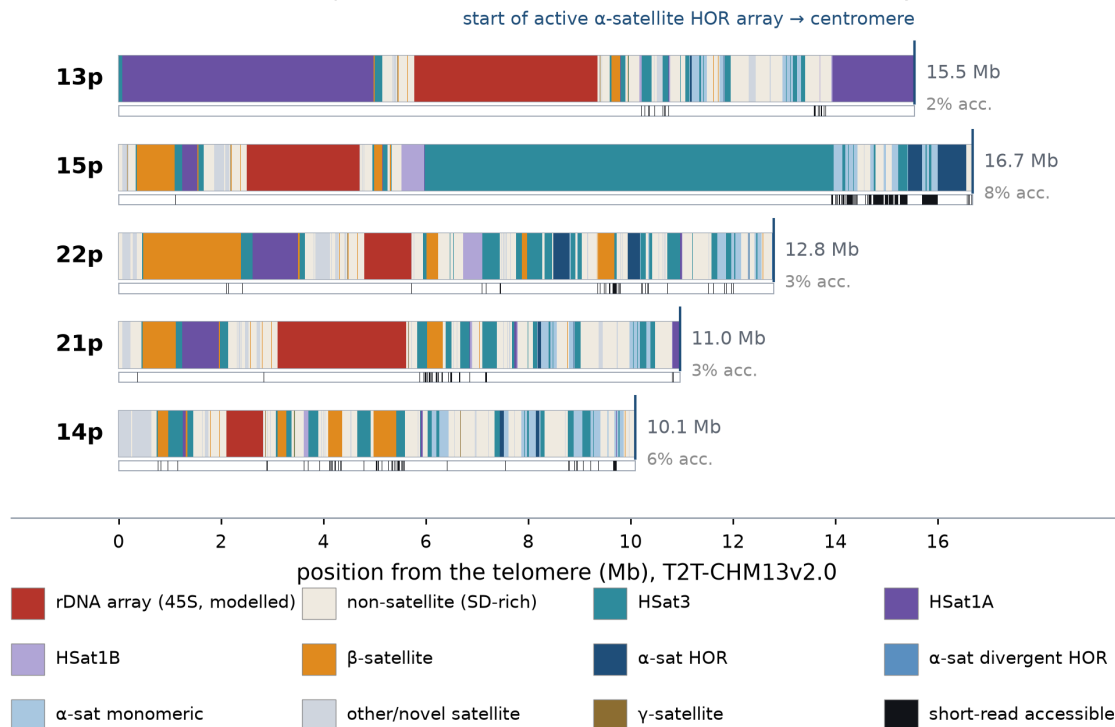
A coordinate caution that has cost real analyses

The five CHM13 rDNA arrays are *models*, not read-supported assemblies: consensus units tiled to approximately the right length. Their coordinates exist, `samtools faidx` will happily return sequence from them, and aligners will map reads into them. None of the resulting numbers mean anything about human rDNA. They must be masked before any coverage, mappability, methylation or variant analysis. The same caution applies, in weaker form, to any satellite array whose length differs between the reference and the sample — which is most of them.

2. A quantitative anatomy

Figure 1 and Table 1 give the measured composition of each arm in T2T-CHM13v2.0. We define the short arm operationally as the interval from the telomere to the start of the *active* α -satellite HOR array (the suprachromosomal family 2 "live" array that recruits the kinetochore), because that boundary is molecularly identifiable, whereas the cytogenetic p11.1/q11.1 border falls inside the centromeric array. On this definition the five arms total 66.07 Mb.

Five short arms, one shared parts list — assembled in a different order on every chromosome



Blocks: CenSat v2.1 satellite annotation. Thin strip beneath each arm: the T2T short-read accessibility mask. Accessible percentage is of the whole arm.

Figure 1. Architecture of the five acrocentric short arms in T2T-CHM13v2.0. Blocks are CenSat v2.1 satellite annotations; the arms are drawn to a common scale from the telomere, and the vertical rule marks the start of the active α -satellite HOR array. The thin strip beneath each arm is the T2T short-read accessibility mask: black indicates sequence in which short-read genotypes are considered reliable. Every arm contains an rDNA array, a distal satellite region and a proximal satellite region, but the identity, order and length of the satellite blocks differ on every chromosome. Note the 4.9 Mb HSat1A block occupying the entire distal third of 13p, and the 8.0 Mb HSat3 block occupying the proximal half of 15p.

Table 1. Measured properties of the five acrocentric short arms. Lengths from the telomere to the start of the active α -satellite HOR array in T2T-CHM13v2.0. GRCh38 non-N bases are counted over the cytogenetic p-arm. rDNA unit counts are derived from modelled array lengths divided by the 44,838 bp reference unit and are therefore properties of the CHM13 model, not of human rDNA copy number. "≥99% twin elsewhere" is the percentage of the arm covered by a segmental duplication whose partner lies in a different acrocentric short arm at ≥99% identity. Accessibility is from the T2T short-read accessibility mask; unique-anchor percentages are from the minimum-unique-*k*-mer-length track.

Property	13p	15p	22p	21p	14p
Short arm (Mb)	15.55	16.68	12.79	10.96	10.09
GRCh38 non-N (Mb)	1.7	2.0	3.68	5.49	1.18
rDNA array (Mb)	3.58	2.2	0.93	2.5	0.72
rDNA units (model)	80	49	21	56	16
Dominant satellite	HSat1A 42%	HSat3 52%	β -sat 21%	HSat3 15%	HSat3 19%
Non-satellite (%)	18.8	10.8	28.9	32.6	38.5
In an SD (%)	58.5	41.5	72.1	87.7	95.7
≥99% twin elsewhere (%)	51.9	22.5	22.5	60.5	44.6
Short-read accessible (%)	1.54	7.8	3.19	2.78	5.63
Unique anchor ≤150 bp (%)	15.6	19.0	43.0	29.9	45.4
Unique anchor ≤10 kb (%)	77.5	85.6	93.9	77.9	93.6
Array-scale only (%)	20.6	11.0	4.8	19.9	4.9
Active α -sat HOR family	S2C13/21H1L	S2C15H1L	S2C14/22H1L	S2C13/21H1L	S2C14/22H1L

Three features of Table 1 deserve emphasis before we proceed to the individual compartments.

First, **the arms are not variations on a theme; they are different objects**. 15p is dominated by a single multi-megabase HSat3 array and has the least non-satellite DNA (10.8%); 14p is 38.5% non-satellite and 95.7% segmentally duplicated; 13p devotes 42% of its length to one HSat1A array and carries the largest rDNA model. A pipeline tuned on one arm will not generalise to the others, and pooling the five arms into a single "acrocentric" stratum obscures more than it reveals.

Second, **the GRCh38 column explains a great deal of published silence**. Where GRCh38 has sequence — 5.49 Mb on 21p, 3.68 Mb on 22p — the literature has content: 21p and 22p appear in older FISH-mapped and BAC-sequenced work far more often than 13p, 14p or 15p. That asymmetry is an artefact of reference availability, not of biology, and it propagates into every resource built on GRCh38 coordinates.

Third, **segmental duplication content is the dominant structural fact**. Against a genome-wide figure of 7.3%, short-arm SD content runs from 41.5% (15p) to 95.7% (14p). More than 60% of the non-rDNA short arm has been described as inter- or intra-chromosomal duplication^{1,7,12}, and our measurement is consistent with that. This is the reason the arms resist analysis, and it is a different reason from the one usually cited. Tandem repetition defeats *assembly*; interchromosomal duplication defeats *assignment*.

3. The compartments, one by one

3.1 Telomere and the distal p13 subtelomeric region

Each short arm terminates in a canonical telomere and a subtelomeric region containing the "ACRO" repeat family — a satellite shared across acrocentric distal ends that serves in practice as the marker confirming that an assembled contig really reaches the distal tip⁵. Distal p13 is otherwise a mosaic of satellite blocks and segmental duplications whose identity differs sharply between arms, and it is the single hardest compartment to assemble correctly: only 41% of short-arm scaffolds in the best-resourced pedigree study to date recovered both distal and proximal sequence, and only 20–40% of distal allelic sequence could be aligned between homologues in the same individual, with most aligned HSat1 showing only ~95% identity⁵. Distal sequence is where allelic divergence is highest, where assembly errors concentrate, and where the "same" region on two homologues may not be recognisably the same region at all.

This has an underappreciated consequence for variant analysis: **on the distal arm, the concept of an allele at a position is frequently undefined**, not because the sequence is hard to read but because there is no correspondence between the two haplotypes to define a position against. Section 13 develops this point.

3.2 The distal junction (DJ)

The DJ is the most important single feature of the acrocentric short arm for anyone trying to obtain a foothold in it. It is a ~300 kb segmental duplication, present in one copy on each of the five arms, immediately distal to the rDNA array and in inverted orientation relative to rDNA transcription¹⁰. It is the most highly conserved sequence shared by all five arms; its discovery and characterisation defined the modern concept of the shared NOR architecture^{10,13}.

Three properties make the DJ analytically central.

It is transcribed and chromatin-marked. The DJ carries promoter and active-transcription chromatin signatures that are conserved across cell types, and produces long non-coding transcripts^{5,10,14}. It is not inert heterochromatin: it is the functional interface between the rDNA array and the rest of the nucleus.

It anchors the NOR to perinucleolar heterochromatin. The DJ is positioned at the nucleolar periphery and is implicated in tethering the NOR within nucleolar architecture, in maintaining the boundary between transcribed rDNA and flanking heterochromatin, and in the reorganisation that follows rDNA damage^{13–14}. Models of nucleolar assembly treat the DJ as the structural element that makes an rDNA array into a NOR.

It is a usable anchor, and it varies. Because the DJ is ~300 kb of non-satellite sequence with chromosome-distinguishing variants, it is the practical handle for identifying which arm a distal contig belongs to. Its total length is remarkably stable — 297–333 kb across a four-generation pedigree — yet a minigraph¹⁵ of twelve error-free DJ sequences resolved into twelve

distinct bubbles, three of them nested, indicating extensive internal structural variation; a chr14 DJ differed from a chr22 DJ by a 1,525 bp deletion and three insertions⁵. Both facts matter: the DJ is conserved enough to be found reliably and variable enough to distinguish arms and haplotypes. **We regard the DJ as the natural coordinate origin for the distal arm** (section 22.1).

3.3 The rDNA array

The rDNA array is the reason the acrocentric short arms exist as a distinct biological category, and the reason they remain unresolved.

Unit structure. The human rDNA repeat unit is 44,838 bp in the reference sequence (GenBank KY962518), of which only 13,332 bp is the 47S pre-rRNA transcription unit: 5' external transcribed spacer, 18S rRNA (1,869 bp), internal transcribed spacer 1, 5.8S rRNA (157 bp), ITS2, 28S rRNA (5,051 bp), 3' ETS. The remaining 31.5 kb — 70% of the unit — is intergenic spacer carrying named tandem-repeat blocks (TR1, TR2), a long repeat (LR1), three simple-sequence-repeat blocks, the spacer promoter and enhancer region, and 26 Alu-family elements making up 12.4% of the unit¹⁶. Figure 4 maps this.

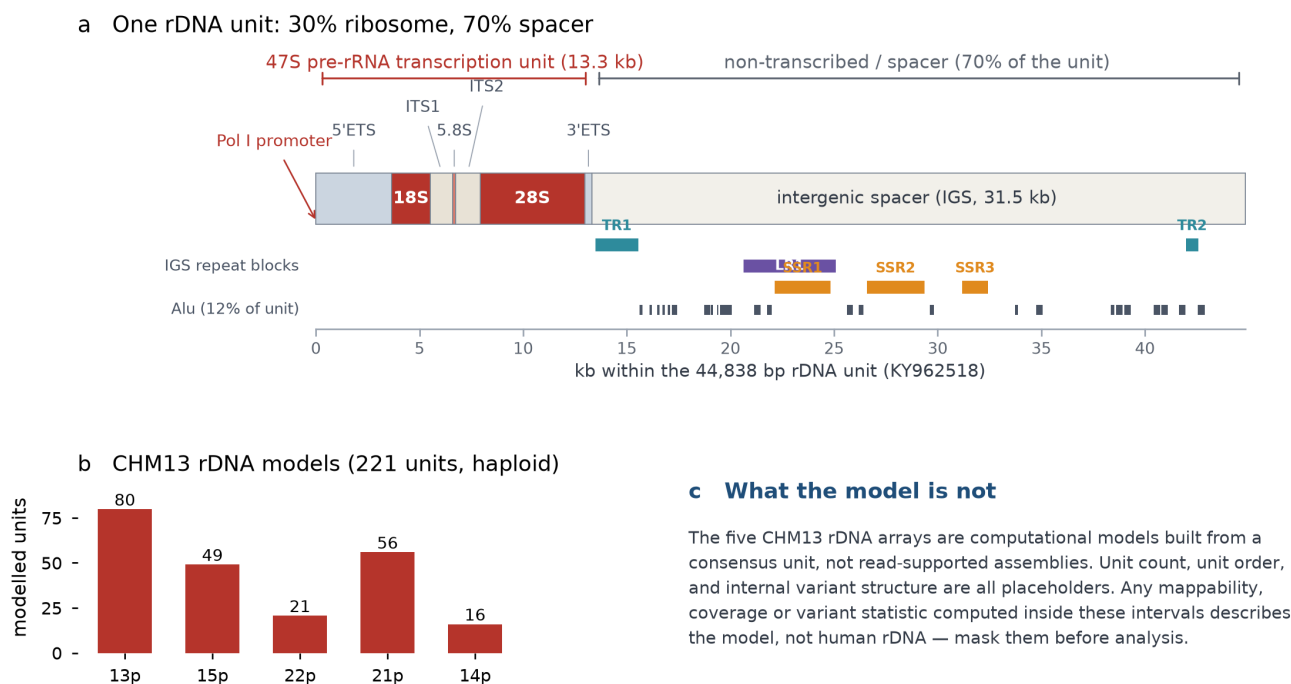


Figure 4. The rDNA unit, the CHM13 rDNA models, and what the models are not. (a) Feature map of the 44,838 bp human rDNA reference unit (GenBank KY962518). Only the leftmost 13.3 kb is transcribed by RNA polymerase I into 47S pre-rRNA; the intergenic spacer carries structured tandem-repeat and simple-sequence-repeat blocks and is dense in Alu elements. (b) Number of units in each CHM13v2.0 rDNA model, obtained by dividing modelled array length by the reference unit length. (c) These arrays are computational models; statistics computed inside them describe the model.

Array organisation. Units are arranged head-to-tail, uniformly oriented with transcription directed centromere-ward, in a single array per arm bounded distally by the DJ and proximally by the PJ^{1,10–11}. Within an array, units are strikingly homogenised — the array behaves as a concerted-evolution unit, with sequence variants shared along a cluster rather than distributed at random^{17–18}. Across arrays on different chromosomes, homogenisation is weaker but still present, which is the sequence signature of exchange between non-homologous arms.

Copy number. The five arrays together typically encode 200–600 rDNA genes per diploid genome, with wide interindividual variation^{1,4,19}. Copy number is coupled to expression and to mitochondrial abundance²⁰, varies in a concerted fashion so that total dosage is partially buffered across arrays²¹, is unstable in cancer and with age^{22–24}, and is heritable in a pattern that reflects array-level rather than base-level inheritance¹⁹. The CHM13 models contain 221 units haploid — within the expected range but not a measurement.

Sequence variation. rDNA units are not identical. Variant positions exist within the transcribed region, are conserved across populations, and are expressed tissue-specifically, implying functionally distinct ribosome populations²⁵. The intergenic spacer is far more variable than the transcribed unit. Long-read work on chromosome 21 rDNA via TAR cloning first showed the extent of unit-to-unit variation directly²⁶, and *ribotin* now reconstructs distinct rDNA "morphs" — recurrent unit haplotypes — from HiFi and ONT data²⁷. Morphs, not units, are the natural allele for rDNA.

Why it does not assemble. Near-identical 45 kb units tiled across 0.7–3.6 Mb defeat every assembler: the repeat period vastly exceeds HiFi read length and approaches the practical limit of ultra-long ONT reads, so the graph contains a single collapsed loop with no way to count traversals. Even in the best current pipelines the rDNA array is a *predictable* collapse — an expected failure, annotated and excluded rather than resolved⁵. Our anchor analysis quantifies this: rDNA is 0.0% short-read accessible, 7.7% anchorable at 10 kb and 16.1% at 100 kb, with a median minimum-unique length of ~565 kb even within the artificially varied model.

3.4 The proximal junction (PJ)

The PJ flanks the rDNA array on the centromeric side, is also shared among arms, and is less well characterised than the DJ — in part because it sits adjacent to, and is partly composed of, satellite sequence, making it harder to isolate and harder to assemble^{10–11}. Functionally it is the second boundary element of the NOR, separating transcribed rDNA from proximal heterochromatin. For analysis it provides a second, weaker anchor into the arm; in practice proximal-arm contigs are more often anchored through the centromere and long arm than through the PJ.

3.5 β -satellite (Sau3A / BSR)

β -satellite is a GC-rich family built on a ~68 bp monomer, organised into higher-order arrays, and it is the satellite most specifically associated with acrocentric short arms — although it also appears at 1q and on the Y²⁸. It totals 6.12 Mb across the five short arms in CHM13 and is the dominant satellite of 22p (2.65 Mb, 21% of the arm). β -satellite is relatively tractable by satellite standards: 7.8% short-read accessible (the highest of any satellite compartment here apart from monomeric α -satellite) and 98.5% anchorable at 10 kb, reflecting substantial internal divergence between and within arrays.

3.6 HSat1A and HSat1B

HSat1 was historically "satellite 1" and is now split into two families with different composition and distribution^{28–29}. HSat1A is extremely AT-rich, built on a ~42 bp repeat, and forms the largest single satellite block on 13p: a 4.9 Mb array spanning the entire distal third of that arm, with 8.79 Mb across all five arms. HSat1B is smaller (1.09 Mb total) and concentrated on 15p and 22p.

Two observations matter analytically. First, HSat1A arrays are **hypervariable in length**: a single chr15 HSat1A array in one pedigree founder reached 1.64 Mb, six times its CHM13 length and 57% of that haplotype's distal sequence, with no assembly error⁵. Second, HSat1A is **mutationally hot** — it carried the largest share of de novo SNVs assigned to any single satellite family, including four in one paternal transmission on 13p, and a 6,763 bp de novo deletion on 22p⁵. The AT-rich composition is a plausible mechanistic contributor: the excess of A>C substitutions on the short arms localises to AT-rich repeat sequence, consistent with oxidative lesions such as 8-hydroxyadenine⁵.

Despite their size, HSat1A arrays are *not* the analytical bottleneck: 98.8% of HSat1A positions have a unique anchor at 10 kb. Divergence within these arrays is high enough that long reads place them. Their difficulty is length variation, not ambiguity.

3.7 HSat2 and HSat3

HSat2 and HSat3 derive from related pentameric and related simple repeats and form the classical C-band heterochromatin of 1q, 9q, 16q and Yq12²⁸. On the acrocentric short arms HSat3 is the single largest satellite compartment — 15.54 Mb, 23.5% of all short-arm sequence — and it is the defining feature of 15p, where an 8.0 Mb block occupies the proximal half of the arm (52.5% of that arm). HSat2 is essentially absent (2 kb).

HSat3 array length is highly variable between haplotypes: 1.69–6.78 Mb on chr15 within one family⁵. A diverged HSat3 block sitting between the main HSat3 array and HSat1B on 15p shares ~93% identity with the main array but is entirely distinct from adjacent HSat1B, and the two blocks differ in CpG methylation (~50% versus ~75%) — evidence that satellite blocks are epigenetically as well as structurally distinguishable⁵. HSat3 transcription is a documented feature of cellular stress and of

cancer, and HSat2/3 RNA is implicated in tumour biology^{29–32}.

3.8 SST1 (NBL2) — the recombinogenic macrosatellite

SST1, also called NBL2, is a ~1.4 kb-unit macrosatellite present on 13p, 14p and 21p (and elsewhere in the genome). It has moved from obscurity to centrality in acrocentric biology over the past three years for two reasons.

It marks the pseudo-homologous regions. Pangenome-graph analysis of HPRC assemblies identified shared homology domains between non-homologous acrocentric arms, centred on SST1, whose positional-homology entropy is consistent with hotspots of ectopic recombination⁴. Long-read assembly of Robertsonian translocation carriers then identified a common breakpoint associated with the SST1 repeat, implicating these sequences directly in the formation of the commonest human constitutional rearrangement³³.

It is epigenetically distinctive and clinically relevant. SST1 arrays are consistently hypermethylated (~78% CpG) relative to flanking sequence (~48%) across 13p, 14p and 21p, and array lengths are bimodally distributed from 21 to 104 kb⁵. Somatic SST1 hypomethylation accompanies tetraploidisation in colorectal cancer³⁴. SST1 is therefore a locus where genotype (array length), epigenotype (methylation) and rearrangement risk converge — an unusually favourable target, discussed in section 22.6.

A caution on mechanism: the single ectopic chr13–chr21 recombination event captured directly in a pedigree did **not** break at SST1. It broke within a 630 kb segmental duplication of 99.53% identity located 1.6 Mb proximal to the SST1 array, at a cluster of PRDM9 binding motifs⁵. The honest current position is that SST1-proximal sequence is a recombination-prone *neighbourhood* containing multiple high-identity substrates, and that SST1 itself may be a marker of that neighbourhood rather than the exclusive substrate.

3.9 α -satellite: monomeric, divergent HOR, and active HOR

α -satellite is organised into higher-order repeats (HORs) built from ~171 bp monomers; an active HOR array recruits CENP-A and the kinetochore, while flanking divergent-HOR and monomeric α -satellite form the pericentromeric transition^{28,35–36}. Across the five short arms (as defined here, i.e. excluding the active array) α -satellite totals 5.54 Mb: 3.77 Mb monomeric, 1.70 Mb HOR and 0.06 Mb divergent HOR, the HOR component being the α -satellite islands within 15p and 22p.

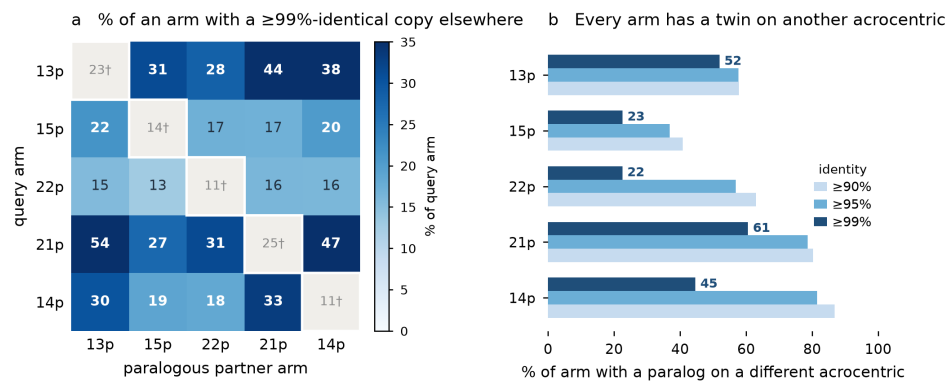
The critical fact for cross-arm analysis is the **identity of the active arrays themselves**:

Chromosome	Active HOR family	Active array in CHM13
13p / 21p	S2C13/21H1L	1.95 Mb / 0.33 Mb
14p / 22p	S2C14/22H1L	2.62 Mb / 2.92 Mb
15p	S2C15H1L	1.02 Mb

Chromosomes 13 and 21 share a centromeric HOR family, and 14 and 22 share a different one. This was known from hybridisation data long before sequence was available³⁷, and it means the paralogy that confounds short-arm analysis extends through the centromere and does not stop at the arm boundary. Reads from a 13 centromere and a 21 centromere are mutually confusable; so are reads from 14 and 22. Centromere length and composition also vary substantially between haplotypes and populations^{38–40}, and chromosome 21 centromere size is asymmetric in families with Down syndrome⁴¹. Monomeric α -satellite is, at 17.6%, the most short-read-accessible compartment of the short arm — the one place where short reads retain partial traction.

3.10 Segmental duplications and the acrocentric "community"

The five short arms should be thought of as a single, partially shared sequence community rather than five independent loci. Figure 5 quantifies this.



SEDEF segmental-duplication calls on T2T-CHM13v2.0, both mates inside a short arm. † diagonal = duplications within the same arm. Satellite arrays add further cross-arm identity beyond these calls.

Figure 5. Each short arm has a near-identical twin on another acrocentric chromosome. (a) Percentage of the query arm (row) covered by a segmental duplication whose partner lies in the labelled arm (column), at ≥99% sequence identity; the diagonal reports duplications internal to the same arm. (b) Percentage of each arm covered by a duplication whose partner lies on a *different* acrocentric chromosome, at three identity thresholds. Calls are SEDEF segmental duplications on T2T-CHM13v2.0, which exclude satellite-dominated intervals; satellite arrays add further cross-arm identity beyond what is shown.

At ≥99% identity, 60.5% of 21p, 51.9% of 13p, 44.6% of 14p and 22.5% each of 15p and 22p lie in a duplication whose partner is a different acrocentric short arm. The pairwise structure recovers the classical cytogenetic pairing preferences: 21p–13p is the strongest relationship (53.6% of 21p has a ≥99% partner in 13p), followed by 21p–14p and 13p–14p; 15p is the most autonomous arm. Crucially, these are *non-satellite* duplications — the SEDEF call set excludes satellite-dominated intervals, and the median satellite fraction of the acrocentric–acrocentric pairs counted here is 0.26. The cross-arm identity contributed by the satellite arrays themselves sits on top of these figures.

This is the structural basis of three phenomena that recur throughout the rest of this report: natural hybrid short arms in the population⁴; Robertsonian translocations^{33,42}; and the intractability of read assignment. Segmental duplications genome-wide show elevated mutation and interlocus gene conversion⁴³; the acrocentric short arms are the extreme case of that phenomenon, and 14 of 103 short-arm de novo SNVs had an identical match elsewhere in the same repeat family, consistent with interlocus gene conversion rather than independent mutation⁵.

4. Function

It is tempting to treat the acrocentric short arms as a genomic dumping ground. They are not. They carry out at least four distinguishable functions, three of which are essential.

4.1 Ribosome biogenesis

The 47S pre-rRNA transcribed from the rDNA arrays supplies the 18S, 5.8S and 28S rRNAs — the catalytic and scaffolding core of every ribosome in the cell. (The 5S rRNA is encoded separately, at 1q42.) RNA polymerase I transcription of rDNA is the single largest transcriptional activity in a growing cell and is regulated by nutrient, growth-factor and stress signalling through a well-characterised machinery centred on UBF, SL1/TIF-IB and the spacer promoter⁴⁴. Not all rDNA units are active: a substantial fraction is held in a silenced, methylated, nucleosome-dense state, and the active/silent ratio is itself regulated. Copy number therefore does not translate directly into output — an important caveat for any attempt to use rDNA dosage as a quantitative phenotype.

4.2 Nucleolar organisation and three-dimensional genome architecture

NORs nucleate nucleoli. Beyond rRNA production, the nucleolus is a hub organising a substantial fraction of the genome: nucleolus-associated domains (NADs) are large, largely heterochromatic regions that sit at the nucleolar periphery, are enriched for satellite DNA and for the acrocentric short arms themselves, and are partially exchangeable with lamina-associated domains between cell cycles^{45–47}. The perinucleolar compartment is a distinct chromatin environment, and the DJ appears to be the element that positions a NOR within it^{13–14}.

Because the five NORs coalesce into one or a few nucleoli, the acrocentric short arms of different chromosomes are brought into physical proximity in essentially every nucleus. Super-resolution imaging shows direct linkages between rDNA on heterologous chromosomes⁴⁸, and this three-dimensional proximity is the standard explanation for why sequence exchange between non-homologous acrocentrics is common^{4,42,49}. It is a rare instance in the human genome where nuclear architecture has a legible, quantifiable consequence in the primary sequence.

The nucleolus is also a stress sensor. Disruption of nucleolar integrity stabilises p53 and triggers cell-cycle arrest⁵⁰, and rDNA damage provokes a localised nucleolar DNA-damage response involving reorganisation of the NOR and its junctions^{13–14}. Inherited defects in ribosome biogenesis produce the ribosomopathies — Diamond-Blackfan anaemia, Treacher Collins syndrome, 5q- syndrome and others — with tissue-restricted phenotypes that are still not fully explained by a general defect in translation⁵¹.

4.3 Centromere function

Each acrocentric carries a functional centromere immediately proximal to the short arm, with an active HOR array, a CENP-A domain, and a hypomethylated **centromere dip region (CDR)** that marks the kinetochore-binding subdomain^{35,38,52}. CDRs on acrocentrics average 242 kb (range 50–480 kb) and are largely, but not perfectly, stable through transmission: 98% of transmitted centromeres show CDR shifts under 200 kb, and 27 of 49 shift by less than 50 kb, with chr14 and chr22 significantly more stable than others; occasional large shifts occur between generations⁵. The functional epigenetic state of the centromere is thus heritable but not fixed — a genuinely unusual mode of inheritance, and one that short-arm-capable assemblies have only just made measurable.

4.4 Satellite transcription and heterochromatin

Satellite arrays are transcribed, and their transcripts are not merely noise. The T2T repeat-element survey documented widespread, compartment-specific transcription and distinct epigenetic states across human satellite families²⁹. HSat2/3 transcripts are strongly induced in stress and in epithelial cancers^{30–32}. Satellite arrays also contribute to chromocentre formation and to the physical clustering of heterochromatin, and acrocentric satellite blocks participate in the perinucleolar heterochromatic shell. Whether specific acrocentric satellite blocks have specific functions, as opposed to generic heterochromatic ones, is open.

4.5 Meiotic behaviour

The acrocentric short arms behave anomalously in meiosis, and this is where function and analysis meet most consequentially.

Direct pedigree measurement found **no allelic meiotic crossover anywhere in 66 Mb of short-arm DNA across 107 transmissions**, while 18 of 19 detected events clustered in the proximal long arm — a 2.3-fold enrichment within 5 Mb of the centromere ($P = 0.0045$) against a genome-wide null, and a significant depletion on the p-arm ($P < 0.0001$)⁵. Fourteen of nineteen events were maternal. Population-scale recombination maps, including the most complete published to date⁵³, exclude these regions by construction, so the pedigree measurement is currently the only direct evidence available. Breakpoints were assessed against PRDM9 binding motifs⁵⁴; the single ectopic event fell 12 kb from the best motif hit⁵. The p-arm depletion is consistent with reduced MLH1 crossover foci on acrocentrics in human oocytes⁵⁵, with an independently observed elevation of crossover rate in acrocentric pericentromeric long-arm regions across 31,228 sperm genomes⁵⁶, and with the observation that synapsis in human male meiosis initiates from a limited number of subtelomeric sites and rarely completes across acrocentric p-arms⁵⁷.

The proposed mechanism is **synapsis failure driven by structural heterozygosity**: homologous short arms in the same individual differ so much in satellite array length and organisation that the synaptonemal complex cannot form along them⁵. This has a cascade of consequences developed in section 21: without synapsis there is no homologous template for repair of meiotic double-strand breaks, which plausibly explains the elevated mutation rate; and crossover demand displaced from the p-arm may load onto the proximal q-arm, where it borders the NAHR-prone regions responsible for the 14q11.2, 15q11.2 and 22q11.2 microdeletion syndromes^{5,58–59}.

5. Variation, population genetics and evolution

5.1 What varies, and by how much

Within a single individual, homologous short arms differ profoundly. In the most detailed assessment available, only 65% of proximal allelic sequence could be aligned between homologues, and only 70% of those aligned bases exceeded 99% identity; on the distal arm only 20–40% of allelic sequence could be aligned at all, with most HSat1 alignments at ~95% identity⁵. By comparison, allelic identity on the long arms is essentially complete. The short arms are the most allelically divergent autosomal sequence in the human genome.

The dominant mode of variation is **array length change**: expansion and contraction of satellite blocks, which shifts everything downstream and destroys positional correspondence. Superimposed on that are unit-level structural variants within HOR arrays, indels within segmental duplications, base substitutions, and — at low frequency — whole-block rearrangements and ectopic exchanges producing hybrid arms.

5.2 Hybrid arms, PHRs and ectopic exchange

Population-scale pangenome analysis found patterns of positional homology between non-homologous acrocentrics indicating that natural "hybrid" short arms — arms carrying a block derived from a different acrocentric chromosome — are present in the general population, and located the implicated pseudo-homologous regions around SST1⁴. Direct pedigree observation put a first bound on the rate: one ectopic chr13–chr21 event in 13 fully assessed transmissions, mediated by a 630 kb, 99.53%-identical duplication 1.6 Mb from SST1⁵. One event in 13 transmissions is a remarkably high per-transmission rate for a structural rearrangement, though the confidence interval is wide and the observation comes from a single family.

5.3 De novo mutation

The short-arm de novo SNV rate is 1.33×10^{-7} per bp per generation — tenfold the autosomal euchromatic rate in the same pedigree (1.37×10^{-8}), fourfold the segmental-duplication rate (2.99×10^{-8}), and comparable to the male-specific region of the Y including Yq12 heterochromatin (1.99×10^{-7}), a region whose complete sequence and population variation are now also resolved and which is similarly satellite-dominated and structurally divergent between individuals^{5,60–61}. Paternal and maternal rates were 2.10×10^{-7} and 0.71×10^{-7} . Eight de novo SVs were detected across 13 transmissions, all in satellite DNA, ranging from 135 bp to 6,763 bp, including stepwise HOR unit gains and losses and a deletion of approximately one SST1 consensus unit⁵. Extrapolating, four to six acrocentric chromosomes change structurally per parental transmission.

The mutational spectrum is also distinctive: CpG>TpG transitions are significantly depleted ($P = 4 \times 10^{-5}$) while C>G ($P = 0.0046$) and A>C ($P = 0.0056$) substitutions are in excess, signatures the authors interpret as mismatch-repair deficiency and oxidative damage respectively⁵. Depleted CpG>TpG has also been observed in other repetitive compartments and may reflect an excess of double-strand breaks with GC-biased gene conversion⁴³.

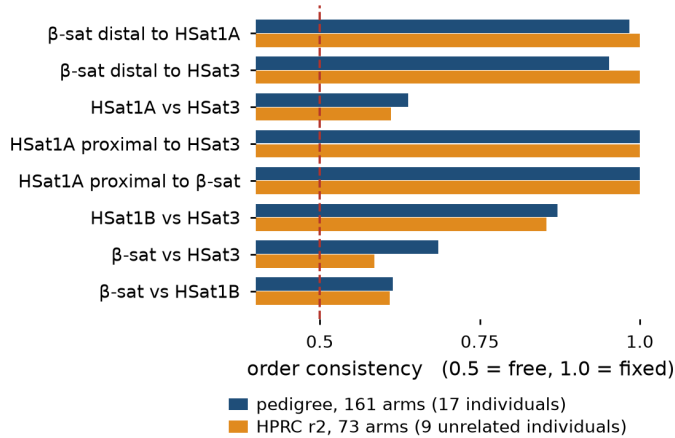
The implication most easily missed

A tenfold elevated SNV rate over 2.1% of the genome means the acrocentric short arms are expected to contribute of order 20% as many de novo SNVs as the entire remaining autosomal euchromatin — from a region that essentially every published DNM catalogue excludes. For a trio study, the short arms are not a rounding error.

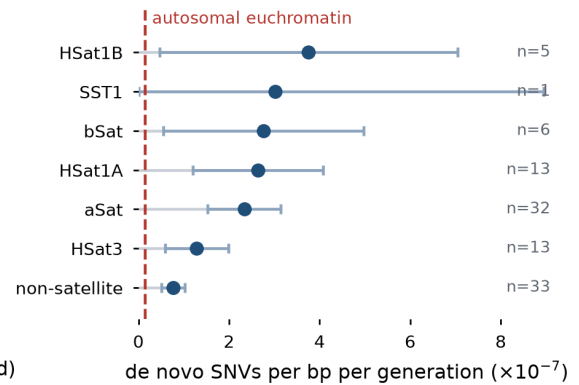
5.4 Where the elevated mutation rate comes from

The headline figure of 1.33×10^{-7} is a single average over a compositionally heterogeneous 66 Mb, and the original study noted that its per-satellite counts were too sparse to claim enrichment for any individual family⁵. The counts can, however, be normalised: we computed callable bases per compartment directly from the pedigree's own satellite annotation over the same region the mutation analysis covered (telomere to the end of the active α -satellite array), removing the sequence flagged by Flagger and NucFreq, and apportioned the study's callable denominator across compartments in that proportion. Methods and caveats in section 25.

a Which block orders are fixed, and which are free



b The tenfold elevation is not uniform



c Position and composition contribute separately

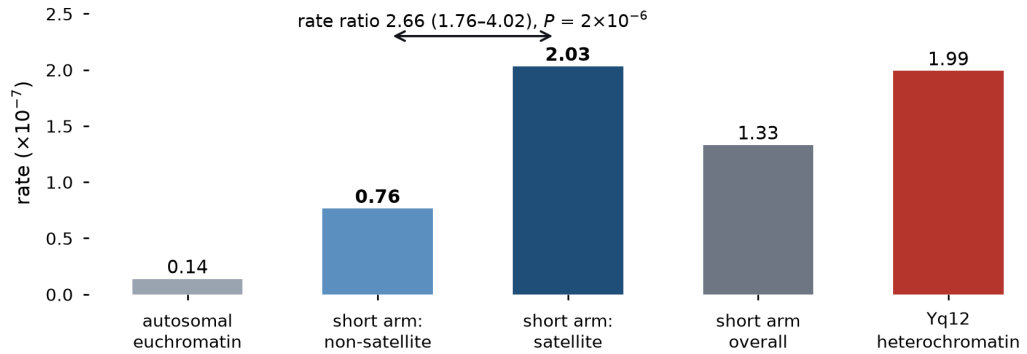


Figure 8. Block-order constraint and the structure of the mutation-rate elevation. (a) Order consistency for eight block relationships, measured independently in the CEPH1463 pedigree and in unrelated HPRC release 2 individuals; 0.5 means the two blocks occur in either order equally often. (b) De novo SNV rate per compartment, obtained by normalising the published per-satellite counts by callable bases computed here; bars are 95% intervals on the count, and the number of observed mutations is given at the right of each row. (c) The elevation decomposes into a position-linked component affecting even non-satellite short-arm sequence and a further composition-linked component specific to satellite DNA. Counts from Lin et al.; denominators and the contrast computed here.

The resulting per-compartment rates span roughly a factor of five, from 0.76×10^{-7} in non-satellite short-arm sequence to $2.3\text{--}3.8 \times 10^{-7}$ in α -satellite, HSat1A, β -satellite and HSat1B (Figure 8b). Individually these are underpowered — most rest on 5–32 mutations and the intervals overlap — and we would not defend a ranking among the satellite families. The aggregate contrast, however, is well powered:

- **satellite sequence:** 70 de novo SNVs in 344.7 Mb of callable sequence = 2.03×10^{-7} per bp per generation, about 15-fold the autosomal euchromatic rate;
- **non-satellite sequence:** 33 de novo SNVs in 431.9 Mb = 0.76×10^{-7} , about 5.6-fold the autosomal rate;
- rate ratio 2.66 (95% CI 1.76–4.02), binomial $P = 2 \times 10^{-6}$.

This decomposition matters because it separates two mechanisms that were proposed together. The original interpretation invoked both a *positional* effect — absence of synapsis and therefore of homologous template for repair — and a *compositional* effect, AT-rich satellite sequence generating oxidative lesions and mismatch-repair-prone contexts⁵. The two make different predictions for non-satellite DNA sitting inside the same recombination desert. A purely compositional model predicts non-satellite short-arm sequence should mutate at roughly the autosomal rate; a purely positional model predicts satellite and non-satellite sequence within the arm should mutate alike. Neither holds. Non-satellite short-arm sequence is elevated 5.6-fold over autosomal euchromatin — considerably more than the 2.2-fold reported for segmental duplications

genome-wide^{5,43}, and these short-arm sequences are largely duplications — while satellite sequence is elevated a further 2.7-fold on top of that. **Both mechanisms appear to operate, with the positional component supplying roughly a fivefold floor across the whole arm and satellite composition roughly tripling it again.**

Three caveats. The denominators are apportioned from compartment composition rather than taken from per-compartment callable tracks, so the absolute per-compartment rates inherit that approximation; the satellite/non-satellite contrast is robust to it provided the apportionment is approximately correct, since it depends only on the ratio of two large denominators. All counts come from a single pedigree. And callability is itself compartment-dependent, so if satellite sequence is systematically harder to call, the satellite rate is if anything an underestimate.

A cleaner test, now specified

The positional floor predicts that non-satellite, non-duplicated sequence elsewhere in a recombination desert should also be elevated, while the same sequence classes in recombining regions should not. The comparison that would settle it is the one already specified in section 21.1 — acrocentric HSat3 against the HSat3 of 1q, 9q and 16q, which is compositionally near-identical but sits in chromosomes that synapse normally. On the numbers here, a positional model predicts the 1q/9q/16q HSat3 rate should fall near $2.03/5.6 \sim 0.4 \times 10^{-7}$, whereas a compositional model predicts it should match the acrocentric satellite figure of $\sim 2 \times 10^{-7}$. That is a fivefold difference between hypotheses, and it is answerable from existing long-read pedigree data with a complete reference.

5.5 Is the order of the blocks the same in everyone?

If the short arms are assembled from a shared parts list (section 2), the natural question is whether the parts are always assembled in the same order. We measured this in two independent datasets: 155 haplotype-resolved short arms annotated for the CEPH1463 pedigree, and 789 chromosome-scale acrocentric arms recovered from 94 unrelated individuals in HPRC release 2 (188 haplotypes, annotated by the independent CenSat pipeline) - 944 arms in total. Because many arms are truncated distally, we used a pairwise statistic that counts only the arms containing both blocks of a pair: for each pair of block types, the fraction of arms in which one is distal to the other, reported as an *order consistency* running from 0.5 (the two blocks occur in either order equally often) to 1.0 (the order never varies). Methods in section 25.

The answer is that **the short arm has a fixed grammar with free slots.**

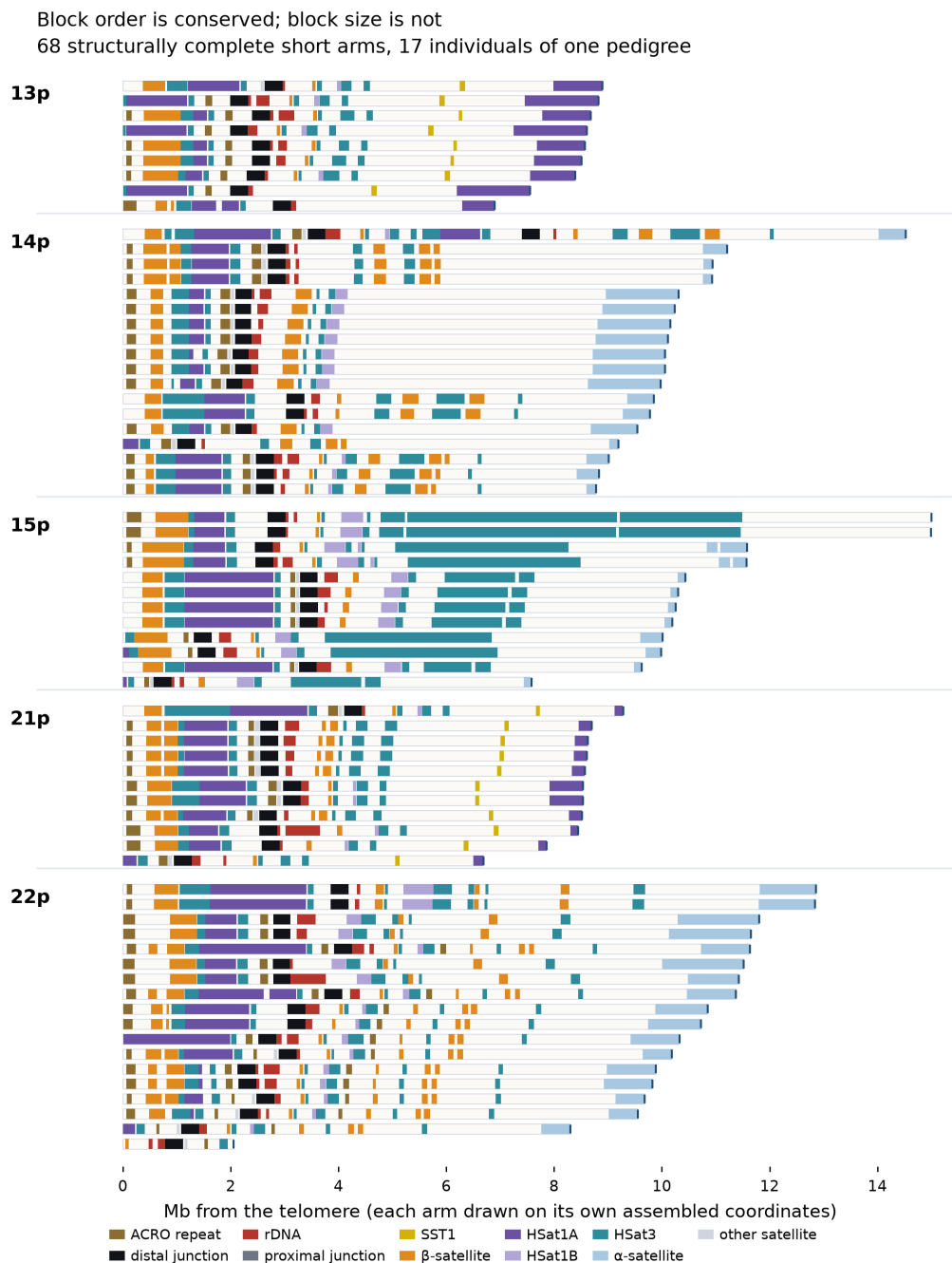


Figure 7. Block order is conserved; block size is not. The 68 structurally complete short arms of the CEPH1463 pedigree, grouped by chromosome and drawn on their own assembled coordinates from the telomere. Figure 9 extends this to 944 arms. Reading left to right, the same block sequence recurs within each chromosome — β -satellite, HSat3, HSat1A, the ACRO repeat, the distal junction, the rDNA array, then a chromosome-specific proximal arrangement — while the position of each boundary shifts by megabases between haplotypes. Arms are ordered by length within each chromosome; the tick at the right end of each arm marks the start of the active α -satellite array.

The landmark backbone is effectively invariant. Across all arms, mean order consistency among ACRO, the DJ, the rDNA array, the PJ and the centromeric α -satellite is 0.969. The rDNA array is proximal to ACRO and the DJ and distal to the centromere in 100% of informative arms ($n = 51\text{--}72$); the DJ is distal to the rDNA in 97.1% ($n = 68$); ACRO is distal to the DJ in 94.7% ($n = 76$). This is the canonical NOR architecture, and on this evidence it is not merely typical but essentially obligate.

Within the two satellite segments, some orders are fixed and others are free. Stratifying relative to the rDNA array removes a confound — HSat1A and HSat3 occur on *both* sides of it, so any whole-arm statistic conflates order with which

copy is larger. Fixed relationships, reproducing in both datasets:

Relationship	Pedigree	HPRC (94 unrelated individuals)
Distal segment: beta-satellite distal to HSat1A	0.980 (n=51)	0.994 (n=171)
Distal segment: beta-satellite distal to HSat3	0.962 (n=52)	0.994 (n=173)
Proximal segment: HSat1A proximal to HSat3	1.000 (n=22)	0.995 (n=214)
Proximal segment: HSat1A proximal to beta-satellite	not informative	1.000 (n=208)
Proximal segment: HSat1B vs HSat3	0.868 (n=68)	0.884 (n=481)
Proximal segment: beta-satellite vs HSat3	not informative	0.860 (n=602)
Distal segment: HSat1A vs HSat3	0.697 (n=66)	0.803 (n=178)
Proximal segment: beta-satellite vs HSat1B	not informative	0.637 (n=463)

So β -satellite always occupies the distal-most satellite slot, HSat1A always occupies the slot immediately abutting the centromere, and SST1 sits at a fixed position immediately distal to that proximal HSat1A block. Most other relationships are strongly but not perfectly constrained (0.80–0.88), and one is close to free: the order of β -satellite and HSat1B in the mid-proximal region is 0.637 over 463 informative arms. Mean consistency is 0.93 for the distal segment and 0.85 for the proximal segment in the unrelated set. The larger dataset revises two estimates upward relative to a nine-individual pilot — β -satellite versus HSat3 and HSat1A versus HSat3 are more ordered than a small sample suggested — which is the expected direction when a consistency statistic is read off few arms.

The distal architecture is shared by all five chromosomes; the proximal architecture is chromosome-specific. Among structurally complete arms, the modal distal satellite order is β -satellite to HSat3 to HSat1A to HSat3 to rDNA on *all five* chromosomes, and most departures from it are truncations rather than rearrangements. The modal proximal orders differ: 13p and 21p terminate β -satellite to HSat3 ($\pm$ HSat1B) to SST1 to HSat1A; 14p and 22p carry an interdigitated β -satellite/HSat3 alternation and no large SST1 array; 15p is dominated by its single multi-megabase HSat3 block. Counting exact satellite-order strings, complete arms fall into 4–6 distinct orders per chromosome (chr13 5 orders in 9 arms, chr14 5 in 18, chr15 6 in 12, chr21 4 in 11, chr22 6 in 18), with a single modal order accounting for a third to a half of each set.

What varies instead is size, by up to an order of magnitude. Pooled across arms, the ratio of the largest to the median per-arm content is 9.5-fold for HSat3 (0.17–7.07 Mb), 5.2-fold for HSat1B, 4.5-fold for the rDNA array, 4.4-fold for α -satellite, 3.6-fold for HSat1A and 2.2-fold for β -satellite. Against that, two elements are almost invariant in length: the DJ (median 0.33 Mb, coefficient of variation 0.00–0.24 within a chromosome) and SST1 (coefficient of variation 0.09–0.16). Our independently derived DJ length distribution — median 0.33 Mb — agrees with the 297–333 kb range reported for this pedigree⁵, which is a useful check on the block-calling procedure.

The short answer

The acrocentric short arms are not built from interchangeable blocks in arbitrary order. There is a conserved syntax — telomere, β -satellite, (HSat1A/HSat3 in either order), ACRO, DJ, rDNA, PJ, a variable HSat3/HSat1B/ β -satellite region, SST1 on 13p/14p/21p, proximal HSat1A, centromere — that is the same on all five chromosomes distal to the rDNA and chromosome-specific proximal to it. Order variation is restricted to a small number of specific slots. Essentially all of the dramatic interindividual difference in these arms is *length* variation within a conserved order, not reshuffling of the order itself.

The practical consequence is encouraging: a conserved block syntax is exactly what makes anchor-relative addressing (section 22.1) and array-level representation (section 22.4) workable. You can label the slots, because the slots are real.

Two limitations. The pedigree arms are related, so some shared order reflects shared descent; the HPRC replication in nine unrelated individuals is what makes the fixed relationships credible, and it reproduces every one that could be tested. And both datasets are limited to arms that assemble — order variation that prevents assembly is, by construction, invisible here.

5.6 Evolution

Acrocentric morphology, NOR number and NOR location differ substantially among great apes, and the complete ape genome assemblies now permit direct comparison of acrocentric architecture across species⁶². Satellite families turn over rapidly, α -satellite HOR families are largely species- and chromosome-specific^{28,38}, and rDNA arrays evolve by concerted evolution

with recurrent gain and loss of entire arrays. Robertsonian fusion is a recurrent mode of karyotype evolution across mammals, and the human recurrent Robertsonians are the same phenomenon caught in progress^{33,42}.

6. Disease

6.1 Robertsonian translocations

Robertsonian (ROB) translocations — whole-arm fusions of two acrocentrics with loss of most short-arm material — are the commonest constitutional structural rearrangement in humans, occurring in roughly 1 in 800 to 1 in 1,000 newborns⁶³. Three pairings dominate: rob(13;14) accounts for the majority, followed by rob(14;21) and rob(13;21). Carriers are usually phenotypically normal but face elevated risks of infertility, recurrent miscarriage, and unbalanced offspring — translocation Down syndrome and trisomy 13 — and rob(14;15) and rob(14;14) carriers are at risk of uniparental disomy for chromosome 14 or 15 in their offspring.

The recurrent pairings are not random, and the sequence basis is now clear: 13, 14 and 21 share the SST1 macrosatellite and the surrounding pseudo-homologous region, and 13–21 and 14–22 share centromeric HOR families^{4,33,37}. Complete sequencing of ROB chromosomes identified a common breakpoint associated with SST1 and clarified how ROB chromosomes form and propagate³³, and a working model places NAHR between high-identity short-arm substrates, brought into proximity by nucleolar clustering, at the centre of the mechanism⁴². Nonhomologous ROB form predominantly in female meiosis⁶⁴.

For diagnostics, the practical point is that **ROB carriers are routinely detected by karyotype and routinely missed by sequencing**. A balanced whole-arm fusion changes almost no unique sequence; copy-number analysis sees nothing; short-read structural-variant callers see nothing. This is a solvable problem and section 22.6 proposes how.

6.2 Aneuploidy, uniparental disomy and imprinting

Trisomy 21, trisomy 13 and trisomy 18 (18 is not acrocentric, but the reasoning about pairing extends) arise predominantly from maternal meiosis I errors, and reduced recombination is an established risk factor for nondisjunction. That the acrocentric p-arms are crossover deserts and the proximal q-arms crossover-enriched is therefore not merely an interesting observation but a candidate contributor to acrocentric-specific aneuploidy risk. Chromosome 21 centromere size asymmetry in Down syndrome families points in the same direction⁴¹.

Trisomy rescue and ROB-mediated mechanisms generate uniparental disomy for chromosomes 14 and 15, producing Temple and Kagami-Ogata syndromes (chr14) and Prader-Willi and Angelman syndromes (chr15)⁶⁵. The 15q11-q13 region additionally hosts recurrent NAHR-mediated deletions, duplications and isodicentric chromosomes, and the 15q11.2 BP1–BP2 microdeletion is a recognised susceptibility locus⁵⁸. Isodicentric chromosome 15 is formed by exchange between the same low-copy repeats, with the acrocentric centromere and short arm as the fusion point. The general mechanistic framework for constitutional rearrangements mediated by repeat architecture applies directly here⁶⁶.

6.3 rDNA in cancer and ageing

rDNA copy number is unstable in cancer: rRNA gene clusters behave as recombinational hotspots in tumours²², copy number is lost and sequence variation increases in several cancers²³, and both amplification and loss occur in a manner linked to tumour genetic context, nucleolar activity and proliferation rate²⁴. Nucleolar size and morphology have been used as prognostic markers in pathology for over a century. rDNA copy number declines with age in some tissues, and rDNA instability is proposed as a general contributor to genome adaptability and to ageing phenotypes⁶⁷. Satellite transcription is aberrantly activated in epithelial cancers and can itself drive genome instability through RNA-derived DNA intermediates^{30–32}, and SST1 hypomethylation accompanies tetraploidisation in colorectal cancer³⁴.

None of these observations has been translated into routine clinical measurement, largely because the measurement is hard. Sections 16 and 22 argue this is now a tractable target rather than an aspiration.

6.4 What is plausibly being missed in rare-disease cohorts

Bringing the preceding sections together, the categories of pathogenic variation that current pediatric and rare-disease pipelines are structurally unable to detect in the acrocentric short arms are:

- balanced Robertsonian translocations and other whole-arm exchanges;
- rDNA copy-number extremes, and rDNA morph composition;
- large satellite array expansions and contractions, including those that alter NOR competence or centromere function;
- de novo SNVs and small SVs in the arms — the 10× enriched class described above;
- ectopic hybrid arms and the recombination-predisposing haplotypes that generate them;
- epigenetic states with genetic consequence, such as CDR position shifts and SST1 or rDNA methylation changes.

Whether any of these causes disease at appreciable frequency is genuinely unknown, because nobody has looked systematically. That is an argument for looking, not for assuming the answer.

Part II · Working with them

References, data generation, assembly, validation, alignment and annotation

7. References and coordinate systems

7.1 The choice of reference is the single largest determinant of what you can see

GRCh38 represents the acrocentric short arms as 66.85 Mb of N across the five cytogenetic p-arms, with 14.05 Mb of real sequence concentrated on 21p and 22p (Figure 2b). Reads originating from short-arm DNA in a GRCh38-aligned sample do not disappear; they mismap — preferentially into the fragments of short-arm sequence that *are* present, into the pericentromeric long arms, and into unplaced scaffolds. This generates spurious coverage peaks, spurious heterozygosity, and a well-documented class of false-positive variant calls that a complete reference removes⁶⁸.

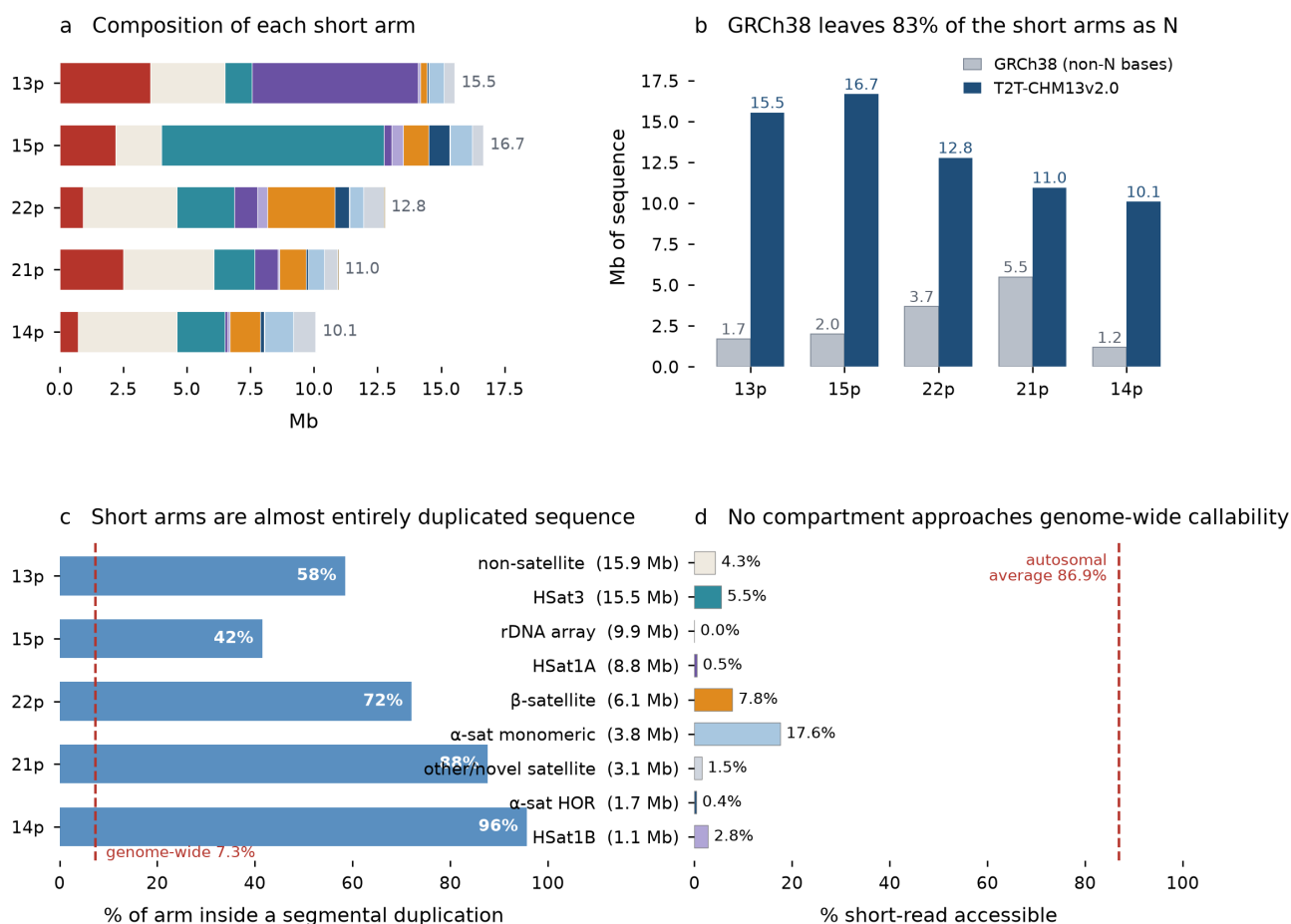


Figure 2. Reference completeness, duplication content and callability. (a) Composition of each short arm by satellite family (colours as Figure 1); the figure at the right of each bar is total arm length in Mb. (b) Sequence actually present in GRCh38 over the cytogenetic p-arm, against the complete T2T-CHM13v2.0 short arm. (c) Percentage of each arm inside a SEDEF segmental duplication, against the genome-wide figure of 7.3%. (d) Percentage of each compartment inside the T2T short-read accessibility mask, against the autosomal average of 86.9%; compartment sizes pooled across the five arms are given in the labels.

Switching to T2T-CHM13 recovers the sequence and, per the T2T analyses, improves variant calling genome-wide^{1,68}. It does not, however, make the short arms callable: our measurement is that 4.27% of short-arm sequence is inside the T2T short-read accessibility mask, against 86.9% of the autosomal genome. **A complete reference converts "invisible" into "visible but unreliable" — which is progress, but only if the unreliability is represented.**

7.2 Practical guidance on reference choice

- **Short-read cohorts already aligned to GRCh38.** Do not re-align an entire cohort solely for the short arms; the yield does not justify it. Instead, add the acrocentric regions to your excluded-region set explicitly (rather than relying on them being N), and treat any variant called within the 14.05 Mb of GRCh38 short-arm sequence as requiring orthogonal confirmation. Re-align if you also want the other T2T benefits — corrected false duplications, medically relevant genes, centromeres.
- **New short-read cohorts.** Align to T2T-CHM13v2.0 (with the HG002 Y for male samples, as distributed) and carry the accessibility mask as a first-class annotation. Report short-arm results only within the mask, and report the masked fraction.
- **Long-read cohorts.** Assemble. Aligning long reads to any single linear reference discards most of the information that made long reads worth generating in these regions; see section 9.
- **Graph references.** A pangenome graph is the correct representation in principle and is increasingly practical^{2,69–71}. Current human pangenome graphs are, however, built from assemblies that are themselves incomplete across the short arms, so the graph inherits the gaps. Graph genotyping^{72–73} works well in the SD-rich non-satellite compartments and poorly inside long arrays. Section 22.2 proposes what an acrocentric-competent graph module would need.

7.3 Liftover and coordinate translation

Liftover between GRCh38 and CHM13 fails in the short arms by construction: there is no source interval to lift from. Even where GRCh38 has sequence, multi-mapping chains make the translation ambiguous. Two rules:

1. **Never lift short-arm coordinates.** Re-derive them by direct alignment or annotation transfer, and record which assembly and which annotation version produced them.
2. **Always record the annotation version.** CenSat v2.0 and v2.1 differ; RepeatMasker versions differ; the rDNA models are versioned. A short-arm coordinate without a reference *and* annotation version is not interpretable.

7.4 Masks you should be carrying

Four T2T-CHM13v2.0 resources should be attached to any short-arm analysis, and all four are used in this report:

Resource	What it gives you	How we used it
CenSat v2.1 satellite annotation	satellite family and array boundaries	composition, compartment stratification
SEDEF segmental duplications (SD.full.bed)	pairwise duplications with identity	cross-arm paralogy (Figure 5)
Short-read accessibility mask	regions where short-read genotypes are trusted	callability (Figures 1, 2d)
Minimum-unique-k-mer-length (MU) tracks	shortest genome-unique substring at each position	anchor ladder, regime map (Figures 3, 6)

The MU tracks in particular are underused. They answer, per base, "how much contiguous sequence do I need to place a read here uniquely?" — which is precisely the question that determines technology choice, and is more informative than a single-read-length mappability track.

8. Data generation: what to buy for what question

Figure 3 is the basis for the recommendations in this section.

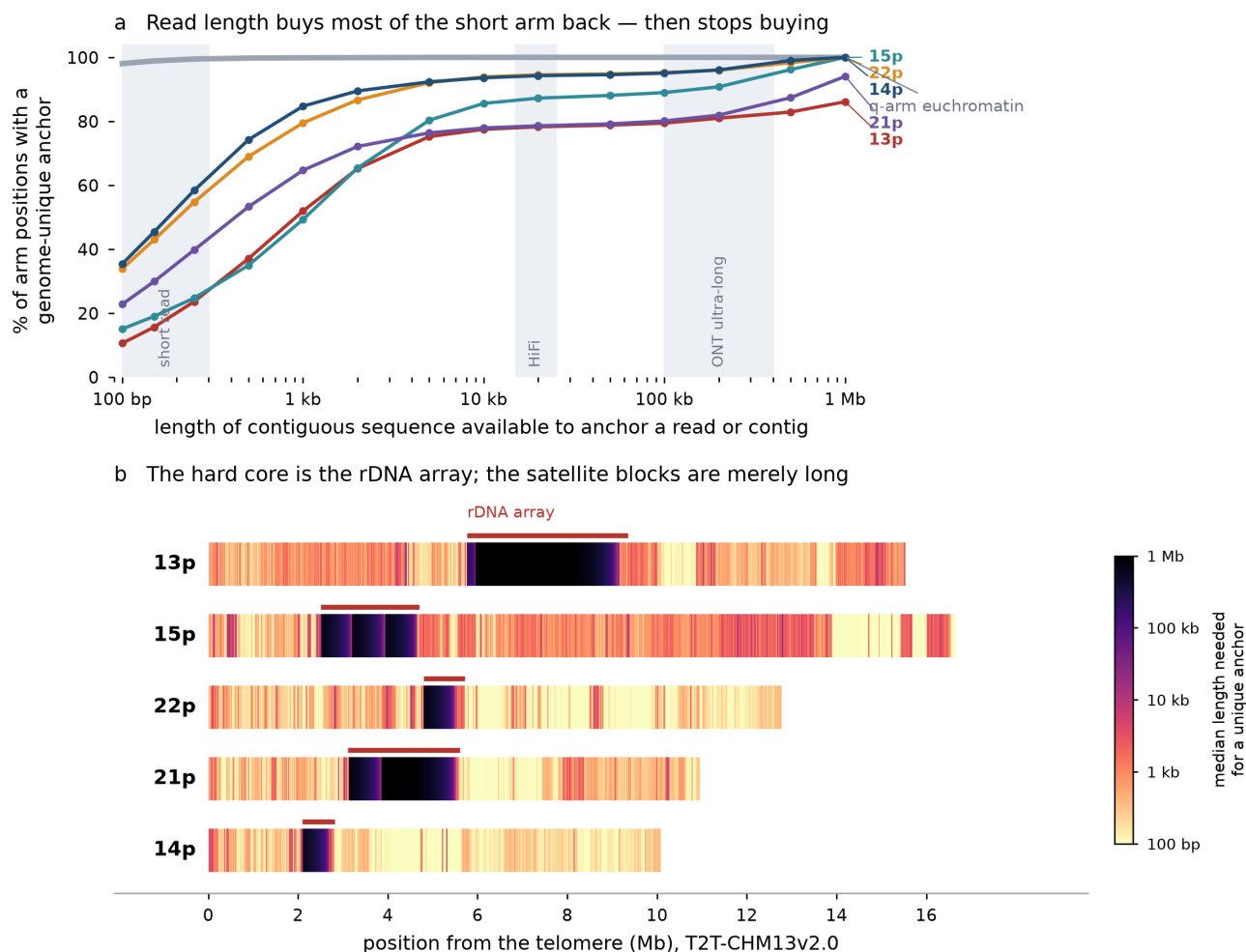


Figure 3. Unique-anchor availability as a function of read length, and where the hard sequence is. (a) Percentage of positions in each short arm at which the shortest genome-unique substring is no longer than the abscissa — equivalently, the fraction of positions at which a read of that length can be placed uniquely in T2T-CHM13v2.0. The grey reference curve is a 20 Mb euchromatic interval of chr13q. Shaded bands mark the working read lengths of current platforms. (b) Median minimum-unique-anchor length in 10 kb windows along each arm; red bars mark the rDNA arrays.

8.1 The read-length ladder, read as a purchasing decision

At 150 bp, 15.6% (13p) to 45.4% (14p) of short-arm positions are uniquely anchorable, against 98.9% in euchromatin. At 1 kb — the scale of linked reads or short-insert mate pairs — this rises to 49.3–84.7%. At 10 kb, the HiFi working range, 77.5–93.6%. Then the curve flattens: from 10 kb to 100 kb, gains are 1.5–3.4 percentage points. From 100 kb to 1 Mb, a second rise appears on 13p (79.4% to 86.1%) and 21p (80.1% to 94.1%) as the rDNA models become traversable.

The purchasing implications are unusually clear:

- **Short reads alone:** adequate for the monomeric α -satellite and parts of β -satellite and the non-satellite compartment, within the accessibility mask; useless for arrays. Do not buy more short-read depth expecting to reach the arms — the limit is uniqueness, not coverage.
- **HiFi (~15–25 kb):** the single highest-value purchase. It captures essentially the whole non-rDNA arm. Depth matters: published short-arm assembly used $\sim 40\times$ HiFi⁵.
- **Ultra-long ONT (>100 kb):** buys little additional *anchoring* but is essential for *assembly* — it spans array-internal repeat periods and resolves graph tangles that HiFi cannot. $\sim 30\times$ UL-ONT with N50 above 100 kb is the working figure^{5,74}.
- **Hi-C / Micro-C (~30 $\times$):** the decisive addition for the distal arm. Proximity ligation is what scaffolds distal to proximal sequence across the unassemblable rDNA array; it raised distal+proximal recovery to 41% of short arms in the best current

study^{5,74}. Without it, distal sequence is usually orphaned.

- **Strand-seq**: independent structural validation and haplotype phasing, particularly for inversions and misorientations^{75–76}.
- **Optical mapping**: still the most direct way to measure multi-megabase array length, and a useful orthogonal check on assembly-derived array sizes.
- **Targeted approaches**: Cas9-guided nanopore enrichment⁷⁷ makes single-locus questions — one rDNA array, one SST1 array — affordable at cohort scale, and is badly underused in this region.
- **Single-molecule chromatin state**: chromatin fibre sequencing resolves regulatory architecture per molecule⁷⁸ and is one of the few ways to ask whether individual rDNA units or satellite blocks are active in a given cell. **Methylation**: obtained free with ONT or HiFi and genuinely informative here, since CDR position, SST1 hypermethylation, satellite block boundaries and rDNA activity are all methylation-visible^{5,52}.

A DNA-extraction caveat that invalidates results

Ultra-long protocols require ultra-high-molecular-weight DNA, and the acrocentric short arms are exactly where fragment length is limiting. Phenol-chloroform or dedicated HMW kits from fresh cells are standard⁵. Equally important: lymphoblastoid cell lines accumulate somatic changes, and satellite arrays and rDNA copy number are among the first things to drift. For mutation or copy-number work, primary tissue is strongly preferable — the pedigree study excluded two generations specifically because only cell lines were available⁵.

8.2 Minimum viable designs

Question	Minimum data	Expected outcome
Is this individual a Robertsonian carrier?	20× HiFi, or karyotype	whole-arm fusion detectable from assembly graph topology or contig structure
rDNA copy number per individual	30× short-read WGS + a calibrated estimator, or ddPCR	total copy number ±10–15%; not per-chromosome
rDNA morph composition	30× HiFi (+ ONT) and ribotin	morph catalogue and relative abundance; not array order
Satellite array lengths, one arm	40× HiFi + 30× UL-ONT	array length to ~kb where the array assembles
Complete phased short arms	40× HiFi + 30× UL-ONT + 30× Hi-C	~90% of expected arm length, rDNA collapsed, ~41% with distal scaffolded
De novo SNVs on short arms	the above, for both parents and child	~30 Mb callable per haplotype; 1–10 SNVs per transmission
Cohort-scale array-length association	short reads + <i>k</i> -mer or depth estimators	relative array-size estimates, cohort-normalised

9. Assembly

9.1 What current assemblers actually achieve

The path to this point ran through single-chromosome proofs of principle⁷⁹, the first complete haploid genome¹, and semi-automated diploid pipelines⁸⁰; the general capabilities and limits of long-read human sequencing are reviewed elsewhere⁸¹.

The honest summary is that **complete short arms are not yet routine, and rDNA arrays do not assemble at all**. Typical diploid long-read assemblies retain 50–100 gaps, and many map to the acrocentric short arms⁸². With 40× HiFi, 30× UL-ONT and 30× Hi-C and a purpose-built assembler, the current state of the art is: ~90% of expected short-arm length recovered per haplotype; rDNA arrays predictably collapsed; 4.8% of bases misassembled or collapsed (~0.72 Mb per haplotype per chromosome); distal and proximal sequence successfully scaffolded together in 41% of arms; 97% of expected haplotypes recovered as pq-scatigs when rDNA and distal sequence are not required⁵.

Plan around these numbers rather than against them. An analysis design that requires complete, error-free short arms will fail; one that requires 30 Mb of *validated callable* sequence per haplotype will succeed.

9.2 Assembler choice

Verkko2 is the current tool of choice for this problem. It builds long-read de Bruijn graphs from HiFi plus ONT and integrates proximity-ligation data for phasing and scaffolding — capabilities added specifically to improve acrocentric short-arm assembly and to handle rare translocations including Robertsonians^{74,83}. **hifiasm** in UL mode is the main alternative and is faster^{84–85}; for centromeres and short arms, published short-arm work has used Verkko2⁵. Polishing with DeepPolisher improves base accuracy⁸⁶; expect QV in the mid-50s⁵.

For the rDNA specifically, **ribotin** is the appropriate tool: rather than attempting a linear assembly, it reconstructs and phases rDNA morphs from reads that map to the rDNA consensus, producing a morph catalogue and abundance estimates²⁷. This is the right shape of answer for the rDNA, and section 22.3 argues it should be generalised to other arrays.

9.3 Identifying and anchoring short-arm contigs

A short-arm contig is worthless unless you can say which chromosome it came from. The established procedure anchors through the long arm^{4–5}:

1. Align assembly contigs to T2T-CHM13 with `minimap2 -x asm20 --secondary=no -s 25000 -c --eqx --cs`⁸⁷.
2. Discard alignments shorter than 100 kb or below 90% identity.
3. Require ≥ 1 Mb of q-arm alignment, at least 1 Mb away from the centromere, so that the anchor is in sequence that segmental duplications do not make ambiguous.
4. Retain the contig as a pq-scatig spanning the p-arm, centromere and 5 Mb of q-arm.
5. Confirm distal completeness by the presence of the subtelomeric ACRO repeat and a DJ flanking the rDNA array.

Steps 3 and 5 are where most errors enter. A p-arm sequence anchored only through a short or centromere-proximal q-arm alignment can be assigned to the wrong chromosome, and 13/21 and 14/22 are the pairs on which this happens.

10. Validation and quality control

Short-arm assemblies fail in characteristic ways, and every one of them produces plausible-looking variant calls. The following are not optional.

Collapse and misassembly detection. NucFreq/NucFlag flags collapsed repeats and misjoins from the frequency of the second-most-common base in read pileups against the assembly — collapse shows up as a region where the second base count exceeds a threshold⁷. HMM-Flagger uses a hidden Markov model on read coverage to classify erroneous, falsely duplicated and collapsed intervals². Use both and take the union as the excluded set; they have different failure modes⁵.

Base-level accuracy. Merqury compares assembly k -mers against short-read k -mers to estimate QV and phasing accuracy without a reference⁸⁸.

Structural visualisation. ModDotPlot for self-similarity dotplots of each arm — the fastest way to see a collapsed array or an inverted misjoin⁸⁹. SVbyEye for all-versus-all comparisons across haplotypes and family members⁹⁰.

Centromere-specific QC. CenMAP identifies centromeric contigs, runs NucFlag, and annotates HOR organisation with RepeatMasker and HumAS-HMMER^{5,35}.

Orthogonal read support. For any variant, require support from two independent long-read technologies in the child and absence in both parental read sets⁵. This is the single most effective false-positive control available in these regions, and it is the reason the pedigree DNM estimate is credible.

General polishing and validation strategy. The T2T consortium's validation and polishing framework remains the reference treatment⁹¹.

The QC step people skip, and why it matters most

Because similar arrays occur on both the distal and proximal sides of the rDNA within the same arm, an assembly that is missing distal sequence will happily align its *proximal* array to the *distal* array of a relative — producing a confident, entirely false, apparent difference. The fix is a 10 kb sliding-window identity check across every parent–offspring alignment, excluding any window below 99.9% identity⁵. Without it, distal/proximal misalignment is a systematic and invisible source of false de novo calls.

11. Alignment, and the assignability problem

11.1 Mappability is not assignability

Every mappability metric, including the MU tracks used here, answers a question about the *reference*: is this substring unique in this assembly? The question that matters for analysis is different: given a read from *this individual*, can it be assigned to the correct locus and the correct haplotype? Call the first *mappability* and the second *assignability*.

In euchromatin the two coincide. In the acrocentric short arms they diverge sharply, for three reasons:

1. **Length variation destroys positional correspondence.** If a sample's HSat1A array is 1.64 Mb and the reference's is 0.27 Mb⁵, reads from the sample's array will map somewhere within the reference array with high confidence and high identity, and the position will be meaningless.
2. **Paralogy is individual-specific.** Which arm a given block is most similar to depends on the individual's haplotypes, including whether they carry a hybrid arm⁴. A read that is uniquely placeable against CHM13 may be genuinely ambiguous against the sample's own genome.
3. **The reference is one haplotype of a hypervariable region.** CHM13 is a single hydatidiform-mole-derived haplotype. For the short arms it is not a consensus in any meaningful sense.

The practical consequence: **MAPQ is not a measure of correctness here.** High MAPQ against CHM13 in a satellite array is frequently high confidence in a wrong answer. Conversely, MAPQ 0 correctly reports ambiguity and should not be filtered away silently — it should be counted and reported.

11.2 Strategies that work

Map to a personalised reference. The highest-value single change. In trio work, align the child's reads to the *parental haplotype assemblies* rather than to a population reference; de novo variants then appear as differences against the actual template from which the child's chromosome was copied⁵. This converts an intractable multi-mapping problem into a tractable two-haplotype problem, and it is what made short-arm DNM measurement possible. The same logic applies outside trios: align to the sample's own assembly.

Use paralog-specific variants (PSVs). Where two paralogous arms must be distinguished, the informative signal is the set of positions at which they differ. Extracting PSVs from a multiple alignment of the candidate paralogs, then genotyping reads at those positions, is the established approach for SD-rich regions and is how the ectopic recombination breakpoint was refined to 18.7 kb^{5,59}.

Align assembly to assembly, not reads to reference. All-versus-all alignment of haplotype assemblies with `minimap2 -x asm20 -c --eqx -D -P --dual=no`, or `wfmash` for larger sets, followed by sliding-window identity computation, is the workhorse comparison in this region^{5,70,87}.

Use graphs where they fit. Pangenome graphs handle the SD-rich non-satellite compartments well and make paralogy explicit rather than implicit^{70–71,92}. They currently handle megabase tandem arrays poorly.

Report the denominator. Whatever strategy is used, the callable denominator must be computed and reported per sample and per arm. An unstated denominator is the commonest defect in short-arm claims.

12. Annotation

Annotation of a new short-arm assembly is a distinct step from assembly and should be done explicitly.

Satellite annotation. The current practical approach maps a curated set of target satellite consensus sequences — derived from CenSat v2.1 plus the HSat consensus set — onto the assembly, inverting the usual query/reference orientation so that all matches to each target are recovered rather than only the best one^{5–6,28}. Alignment blocks within 500 bp are merged and filtered by length to suppress spurious LINE matches. This is a pragmatic procedure, not a principled classifier, and the field would benefit from a proper probabilistic satellite annotator (section 20.9).

α -satellite HOR structure. HumAS-HMMER/HumAS-SD for HOR monomer classification, integrated in CenMAP^{5,35}.

Repeats and masking. RepeatMasker, Tandem Repeats Finder and WindowMasker in combination, with the union used as the soft-mask, before SD detection^{5,93–94}.

Segmental duplications. SEDEF on the repeat-soft-masked assembly, retaining duplications above 1 kb and 90% identity; the conventional additional filter excludes duplications more than 70% satellite^{5,95}. Be aware that this filter is what makes SD statistics and satellite statistics non-overlapping; it also means published SD content *understates* total paralogy.

Landmarks. Telomeres and gaps with seqtk `telo/gap`; DJ and PJ by alignment to reference junction sequences; rDNA by alignment to the KY962518 unit, rotated to begin upstream of the 45S transcription start site so that the promoter is included^{5,16}.

Methylation. Basecall with modification tags, align to the sample's own assembly, and pile up with modkit; 10 kb bins are a reasonable default for array-scale comparison⁵.

Part III · Variants, from single nucleotides to whole arms

A framework, then class-by-class practice

13. A three-regime framework, and why VCF breaks

13.1 The framework

Compartment labels (HSat3, β -satellite, rDNA) describe sequence content. They are not the right axis for deciding how to analyse a region, because compartments with the same name behave differently depending on array length and internal divergence — HSat1A is a satellite and is 98.8% anchorable at 10 kb; rDNA is a satellite and is 7.7% anchorable at 10 kb. The right axis is **how much contiguous sequence is needed to place a read uniquely**, which is exactly what the MU tracks measure. Figure 6 partitions the short arms on that axis into three regimes.

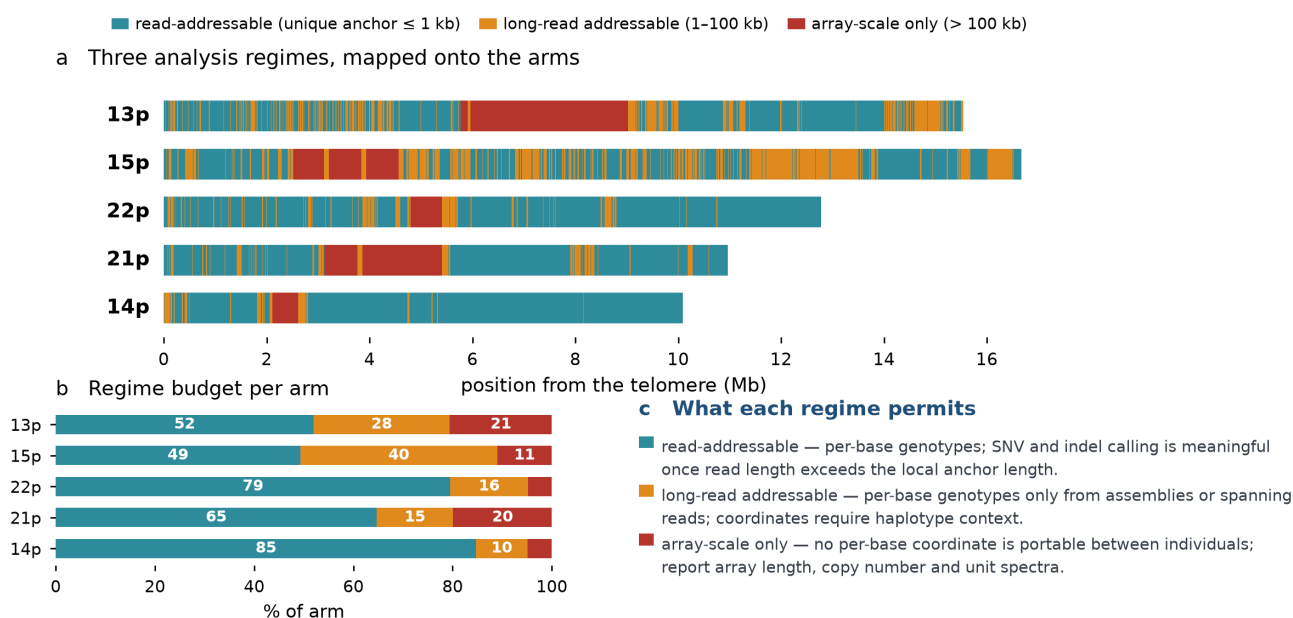


Figure 6. Three analysis regimes across the acrocentric short arms. (a) Every 10 kb window of each arm classified by its median minimum-unique-anchor length. (b) The proportion of each arm falling in each regime. (c) What each regime permits analytically. Regime boundaries at 1 kb and 100 kb correspond approximately to the transition from short-read to long-read addressability, and from long-read to array-scale-only.

Regime I — read-addressable (unique anchor ≤ 1 kb). 49–85% of each arm. Per-base genotypes are meaningful. Short reads recover part of this regime (the part with anchors under ~ 150 bp, plus paired-end reach); HiFi recovers essentially all of it. Standard SNV and small-indel calling applies, subject to the caveats in section 14.

Regime II — long-read addressable (1–100 kb). 10–40% of each arm; 39.7% of 15p. Per-base genotypes are obtainable, but only from assemblies or from reads that span the local repeat period, and the coordinates require haplotype context to be interpretable. Standard read-to-reference pipelines produce confident wrong answers here.

Regime III — array-scale only (> 100 kb). 4.8–20.6% of each arm; 20.6% of 13p and 19.9% of 21p. No per-base coordinate in this regime is portable between individuals. The meaningful observables are array-level: length, copy number, unit composition, methylation state.

13.2 Why the VCF model fails in Regimes II and III

A VCF record asserts that at position p of reference R , this sample carries allele A instead of the reference allele. That assertion presupposes that position p has a counterpart in the sample's genome. Inside a satellite array whose length differs between

reference and sample by hundreds of kilobases — the normal case — there is no such counterpart. The aligner will nonetheless produce a position, and the caller will nonetheless produce a record, and the record will be reproducible, and it will still be meaningless: it describes the alignment's arbitrary choice of where to place the indel burden, not a property of the sample.

This is not a pedantic objection. It has three concrete consequences that we have seen repeatedly in practice and that are visible in the numbers above:

1. **Non-comparability across samples.** Two samples with identical arrays but different alignment paths get different VCF records. Two samples with genuinely different arrays may get the same record.
2. **Non-comparability across references.** The same sample re-aligned to a different assembly, or to its own assembly, yields a different record set — with no mapping between them.
3. **Accumulating false discovery in cohort analysis.** Because the errors are reference-driven, they are *correlated across samples* in ways that mimic real population structure. Any association test run over Regime II/III VCF records is testing alignment behaviour.

Our position is that **per-base variant representation should be abandoned in Regime III and used only with haplotype context in Regime II**. Section 22.4 sketches what should replace it.

13.3 The decision matrix

Table 6 maps variant class against regime to a recommended method. It is the operational summary of this report.

Variant class	Regime I	Regime II	Regime III
SNV / small indel	standard calling; require accessibility mask for short reads; HiFi assembly-based calls preferred	assembly-to-assembly calling against a personalised reference; orthogonal two-technology read support mandatory	not attempted; report as not assessed
Indel 50 bp – 10 kb	long-read SV calling (sniffles2, delly), assembly confirmation	assembly-to-assembly with Truvari matching at 80% size/sequence; manual review	not attempted
Tandem-repeat unit change (e.g. HOR <i>n</i>-mer gain/loss)	tandem-repeat genotyping (TRGT) where the array is short and anchored	assembly-based unit-structure comparison; HOR annotation diff	array-level unit-spectrum comparison only
Satellite array length	direct from assembly	direct from assembly where the array assembles; otherwise optical mapping or read-depth	read-depth or <i>k</i> -mer estimate; optical mapping; no coordinates
rDNA copy number	—	—	calibrated read-depth, <i>k</i> -mer, or ddPCR; total and not per-chromosome
rDNA sequence variation	—	—	morph reconstruction (ribotin); report morphs and abundances, not positions
Large SV / NAHR (> 100 kb)	assembly all-vs-all; PSV-based breakpoint refinement	same, plus Strand-seq and Hi-C confirmation	inferred from flanking Regime I/II anchors; array content not resolvable
Whole-arm exchange (Robertsonian, hybrid arm)	assembly graph topology; contig-level pq structure; karyotype remains valid	same	same
Methylation / CDR state	standard modified-basecall pileup	same, on personalised reference	array-level mean methylation; CDR position relative to array start

14. SNVs and indels, including de novo calling in trios

This section is written for the case we expect to be most common: a trio or pedigree cohort in which the short arms have been excluded, and the question is what can be recovered.

14.1 What is callable, and how to define it

Three nested denominators must be distinguished, and reported:

- **Reference sequence present:** 66.07 Mb (T2T-CHM13), 14.05 Mb (GRCh38).
- **Short-read accessible:** 2.82 Mb — 4.27% of the short arms. This is the honest short-read denominator.
- **Long-read/assembly callable:** ~30.7 Mb per haplotype in the best current pedigree analysis, after excluding rDNA, assembly-error regions flagged by NucFreq and Flagger, misaligned windows below 99.9% identity, and unanchored distal sequence⁵.

The order-of-magnitude difference between the second and third figures is the entire argument for long reads in this region.

The callable-region definition that we recommend adopting, following⁵, is the intersection of: (i) transmitted/aligned blocks ≥ 1 Mb at $\geq 99\%$ identity to the template haplotype; (ii) outside the union of NucFreq and Flagger flags on *both* query and reference; (iii) outside any 10 kb window below 99.9% identity between parent and child; (iv) outside rDNA models and unresolved arrays. Report it per sample, per haplotype, per chromosome.

14.2 De novo SNV calling: the procedure that works

The assembly-first, personalised-reference procedure is currently the only one with demonstrated validity in these regions⁵:

1. Assemble and phase both parents and the child (40× HiFi, 30× UL-ONT, 30× Hi-C each); anchor short arms as pq-scatis.
2. Align child haplotypes to parental haplotypes (`wfmash -s 50k -l 150k -p 90 -n 1`), keeping blocks >1 Mb at $>99\%$ identity as the transmitted set.
3. Call variants from the haplotype-to-haplotype alignment (`wgtools call`)⁹⁶.
4. Require every candidate to be outside assembly-error regions on both sequences.
5. Require ≥ 3 supporting child reads in *both* HiFi and UL-ONT, and absence from both parental HiFi and UL-ONT read sets.
6. Manually review every candidate.
7. Where a further generation exists, use transmission as validation — 85% of G3 candidates were confirmed as transmitted⁵.

Expect 1–10 de novo SNVs per transmission per individual across all five arms; for comparison, familial long-read sequencing raises de novo yield genome-wide relative to short-read trio calling⁹⁷; ~80% will fall in annotated satellite or SD sequence⁵.

14.3 What fails, and the specific false-positive classes

- **Short-read trio DNM calling in the short arms is not salvageable.** With 4.27% accessibility and pervasive paralogy, the false-positive rate from mismapping exceeds any plausible true rate. If you have only short reads, the correct action is to exclude the short arms *explicitly and with a stated denominator*, not to filter aggressively and hope.
- **Distal/proximal misalignment** produces false de novo calls with full read support on both sides, because the reads genuinely support the sequence — just at the wrong locus. Only the sliding-window identity check catches it.
- **Collapsed arrays** produce apparent heterozygosity and apparent de novo variants at high read support. NucFreq is the control.
- **Interlocus gene conversion masquerades as de novo mutation.** Fourteen of 103 short-arm de novo SNVs matched an existing allele elsewhere in the same repeat family⁵. These are real sequence changes but the mechanism is conversion, not point mutation, and they should be annotated separately: check every candidate against the set of alleles present elsewhere in the same repeat family in the same individual.
- **Cell-line artefacts.** Satellite and rDNA changes accumulate in culture. Use primary material for mutation work.

14.4 Recalibrating expectations

If you do recover short-arm DNMs, the expected rate is $\sim 10\times$ the autosomal rate, with depleted CpG>TpG and elevated C>G and A>C⁵. Two practical consequences: a short-arm DNM count that looks like the autosomal rate is probably an underestimate reflecting incomplete ascertainment, not a negative result; and a mutational-signature analysis that pools short-arm and euchromatic DNMs will be distorted by the short-arm spectrum.

15. Tandem-repeat and array-level variation

Array-level variation is the dominant mode of variation in these regions, and it is the class least well served by existing tooling.

What to measure. Array length; unit copy number; unit-type composition (for HOR arrays, the n -mer spectrum); internal divergence; boundary positions relative to flanking anchors; methylation profile.

From assemblies. Where an array assembles cleanly, measure it directly and report the flanking anchor sequences used to delimit it. Validate with NucFreq — a collapsed array reports a spuriously short length, which is the commonest error in published array sizes.

Unit-level SVs. De novo unit gains and losses in α -satellite HOR arrays are detectable and real: stepwise deletion of a 1,364 bp 8-mer on chr22 in two independent transmissions, and insertion of a 680 bp 4-mer on chr21⁵. These are best represented as changes in the HOR unit spectrum rather than as indels at coordinates. TRGT is the appropriate genotyper where arrays are short and anchorable⁹⁸; the broader population context for tandem-repeat variation and its expression consequences provides the comparison baseline⁹⁹.

From short reads. Relative array length can be estimated from read depth over array-specific k -mers, with cohort-internal normalisation. This is a *relative* measure, useful for cohort comparisons and association testing, and it does not yield per-chromosome resolution for families shared between arms. Validate against a subset with long-read assemblies.

Report array length as a quantitative trait, with units, the estimator, and the normalisation. This is the representation that survives comparison across studies; a VCF indel record is not.

16. rDNA copy number and sequence

The rDNA deserves separate treatment because it is simultaneously the most biologically consequential and least tractable compartment.

Copy number. Diploid totals are typically 200–600 units with wide variation^{1,19}. Estimation options, in decreasing order of practicality:

- *Short-read depth*: coverage over the rDNA unit relative to genome-wide coverage, with GC and mappability correction. Cheap, cohort-scalable, sensitive to library preparation and to the reference unit used. Requires within-cohort normalisation and, ideally, calibration against an orthogonal method on a subset. Note that the CHM13 rDNA models must be masked; use the KY962518 unit as the target instead.
- *ddPCR*: the accuracy standard for total copy number; low throughput.
- *k-mer based*: count rDNA-specific k -mers in raw reads; avoids alignment entirely and is robust, and this is the approach we would recommend for large cohorts.
- *Long-read depth over collapsed arrays*: usable, but long-read depth is less uniform.
- *Optical mapping / FISH*: per-chromosome array size, low throughput, and the only routes to chromosome-resolved counts.

Per-chromosome assignment is not currently solvable at scale, and claims of per-chromosome rDNA copy number from short reads should be treated sceptically. Chromosome-specific IGS variants exist and are the plausible basis for a solution (section 22.5).

Sequence variation and morphs. Use *ribotin* to reconstruct morphs from HiFi (+ONT)²⁷. Report the morph catalogue and abundances. Transcribed-region variants are functionally interesting and tissue-specifically expressed²⁵; IGS variants are more numerous and are the better barcodes. Do not report rDNA variants as VCF records against a single consensus unit: the same variant in a different unit is a different allele, and abundance rather than genotype is the meaningful quantity.

Methylation and activity. The active/silent unit ratio is regulated and is not inferable from copy number⁴⁴. Long-read methylation over rDNA provides a proxy and should be reported alongside copy number in any study proposing rDNA dosage as a phenotype.

17. Large structural variants, NAHR and whole-arm events

Ectopic exchange and hybrid arms. Detection requires assemblies and all-versus-all comparison; the diagnostic signature is a contig whose alignment transitions from one chromosome's haplotype to another's at a defined point, with read coverage from the sample dropping at the breakpoint on one template and resuming on the other⁵. Breakpoint refinement uses paralog-specific variants in a multiple alignment of the candidate paralogs, reaching ~18 kb resolution in the published case. Confirm with Strand-seq and, where available, Hi-C contact patterns.

Robertsonian translocations. Complete sequencing is now possible and has been done³³; Verkko2 explicitly supports translocation assembly⁷⁴. For detection at scale, the assembly graph is more informative than any alignment: a whole-arm fusion appears as a contig carrying two different long arms joined through acrocentric short-arm/centromeric sequence. Karyotype remains a valid and cheap orthogonal test and should not be abandoned.

Inversions. Pericentromeric inversion polymorphisms are common on acrocentric long arms — a 515 kb 14q11 inversion segregates in the CEPH1463 pedigree, and q-arm inversion polymorphisms were found on chr14, chr15 and chr22^{5,76}. They matter for short-arm work because they delimit recombination and can disrupt synapsis; the three chr15 crossovers observed in that pedigree all mapped immediately adjacent to an SD-mediated inversion⁵. Strand-seq is the method of choice⁷⁵.

Isodicentrics and marker chromosomes. Acrocentric-derived small supernumerary marker chromosomes are a recurrent clinical finding, usually characterised cytogenetically. Sequence-level characterisation is now feasible and largely undone.

18. Somatic and cancer applications

Three observables are within reach and largely unexploited.

rDNA copy-number change in tumours. Measurable from existing short-read tumour/normal WGS with a calibrated estimator, at no additional sequencing cost. Given the established instability of rDNA in cancer^{22–24}, this is an obvious retrospective analysis on existing cohorts.

Satellite transcription and hypomethylation. HSat2/3 transcription and SST1 hypomethylation are established cancer-associated changes^{30–32,34}. Both are measurable from data types that large tumour cohorts already have, provided a complete reference is used.

Mosaic and clonal structural change. Acrocentric whole-arm events and satellite array changes in somatic tissues are essentially uncharacterised. Sensitive mosaic SV detection is available for long reads¹⁰⁰, and read-based callers and integrated pipelines developed for short and long reads^{101–103} can be applied within the anchorable fraction and, applied to acrocentrics, would answer whether somatic acrocentric instability is a real phenomenon at appreciable frequency.

19. Short-read cohorts: what is actually possible

Most existing cohorts — including most rare-disease and pediatric cohorts — are short-read. The following is what we consider defensible, and what we consider not.

Defensible.

1. **Explicit exclusion with a stated denominator.** Define the acrocentric short arms in your reference's coordinates, exclude them from variant analysis, and state the excluded base count in methods. This is strictly better than relying on N-masking, which silently redistributes the reads.
2. **Accessibility-masked calling on T2T.** Within the 2.82 Mb accessible fraction, short-read genotypes are usable. Report results only there, with the mask cited.
3. **Total rDNA copy number as a quantitative trait.** Cohort-normalised, *k*-mer or depth-based, validated on a long-read subset. This is a real, heritable, biologically consequential phenotype^{20–21} and it is obtainable from data you already have.

4. **Relative satellite array-size estimates.** Family-specific k -mer depth for HSat1A, HSat3, β -satellite and SST1, cohort-normalised. Useful as quantitative traits and for outlier detection.
5. **Robertsonian carrier screening by an unusual-coverage signature.** A ROB carrier has lost most of two short arms; the resulting deficit in short-arm satellite and rDNA k -mer depth is a candidate screening signal. This needs validation against known carriers (section 22.6), but the data requirement is nil.
6. **Outlier detection rather than genotyping.** Even where per-base genotypes are meaningless, gross per-sample summaries (total satellite k -mer content per family, rDNA copy number, arm-level coverage profile) are comparable within a cohort and will flag the extreme individuals who merit long-read follow-up. We regard this as the highest-value short-read strategy available.

Not defensible.

- Per-base SNV or indel calls outside the accessibility mask.
- De novo mutation calls anywhere in the short arms from short reads alone.
- Per-chromosome rDNA copy number.
- Structural-variant calls inside satellite arrays.
- Any claim of *absence* of variation in these regions. With 4.27% accessibility, absence of calls carries essentially no information — and this is the most frequent unforced error in the literature. The correct phrasing is "not assessed", not "no variants found".

Part IV · Challenges, proposals and an agenda

What is unresolved, what we propose doing about it, and what we think is true but unproven

20. Twelve unresolved challenges

These are ordered by how much we think solving them would unlock, not by difficulty.

20.1 There is no portable coordinate system. Every short-arm coordinate is a statement about one assembly. Because array lengths differ between haplotypes by megabases, coordinates do not transfer between individuals even in principle. This blocks cross-study comparison, database curation, clinical reporting and meta-analysis simultaneously. Addressed in 22.1.

20.2 There is no truth set. There is no GIAB-style benchmark for the acrocentric short arms — no truth variants, no confident regions, no stratifications. Consequently no caller has ever been evaluated here, and every performance claim in this region is unverified. The medically-relevant-gene benchmarks deliberately stopped short of these regions^{104–105}. Addressed in 22.7.

20.3 The rDNA arrays remain unassembled, and no current technology will assemble them. A 45 kb near-identical unit tiled over up to 3.6 Mb cannot be resolved by 20 kb or 100 kb reads. Morph reconstruction²⁷ gives the unit spectrum but not the array order or an exact count. Solving this requires either reads approaching megabase length, single-molecule physical mapping of whole arrays, or an in-situ approach.

20.4 Assignability is unmeasured. We have mappability tracks for the reference. We have no metric for "probability that a read from this individual is assigned to the correct arm and haplotype", and therefore no principled way to set a confidence threshold. Addressed in 22.3.

20.5 Population-scale short-arm data do not exist. HPRC and HGSC assemblies are the best available and are incomplete across the short arms^{2–3,82}; the deepest short-arm dataset is a single pedigree of 16 informative samples⁵. Allele frequencies, array-length distributions, hybrid-arm frequencies and morph frequencies are effectively unknown at population scale. Nothing in this region can be interpreted clinically without that denominator.

20.6 Satellite annotation is heuristic. Current practice maps consensus sequences and merges hits with length filters⁵. There is no probabilistic classifier, no calibrated confidence, and no agreed taxonomy boundary between families (the "diverged HSat3" block on 15p at 93% identity to the main array is a case in point). Different groups will annotate the same assembly differently.

20.7 Array-level variation has no data model. There is no standard format, no exchange representation, no database schema for "this haplotype's HSat3 array is 4.2 Mb with this unit spectrum". Addressed in 22.4.

20.8 Statistical genetics for array traits is undeveloped. Array lengths are quantitative, highly mutable, non-biallelic and measured with substantial error. Standard GWAS and burden machinery does not apply. Mutation rates high enough to produce recurrent identical-by-state alleles break the assumptions of identity-by-descent methods.

20.9 The functional link is missing. We can now measure short-arm genotype. We almost never measure the corresponding phenotype — rRNA output, nucleolar number and morphology, ribosome composition, kinetochore function. Without paired functional readouts, short-arm variation cannot be interpreted.

20.10 Clinical translation has not begun. No clinical laboratory reports acrocentric short-arm variation from sequencing. Robertsonian carriers are detected by karyotype, an assay from 1959, because no sequencing-based test has been validated. Addressed in 22.6.

20.11 Reference panels are built from cell lines. Satellite arrays and rDNA drift in culture, and the community's reference materials — including much of HPRC — are lymphoblastoid lines. The extent to which current "reference" short-arm haplotypes reflect primary genomes is unknown and should be measured.

20.12 Cost and compute. Complete short arms currently require three sequencing technologies per sample plus substantial compute. Until the requirement falls to one technology, population-scale short-arm genomics will not happen.

21. Connections we think are real but unproven

The following are inferences and hypotheses, not findings. We state each with the prediction that would test it, so that they can be disposed of efficiently.

21.1 Synopsis failure as a general mutational mechanism

The observed coincidence on the acrocentric short arms — no allelic crossover, tenfold elevated SNV rate, mismatch-repair-like and oxidative signatures — has been interpreted as causal: without synapsis there is no homologous template, so meiotic double-strand breaks and mismatches are repaired by lower-fidelity routes⁵. If that mechanism is general, it predicts a genome-wide inverse relationship between local synapsis competence and de novo mutation rate, independent of sequence composition.

The decomposition in section 5.4 is a first partial test and is consistent with the positional model supplying a floor: non-satellite sequence inside the arm mutates 5.6-fold faster than autosomal euchromatin, which composition alone does not explain.

Predictions. (i) Regions heterozygous for large inversions should show locally elevated DNM rates in the heterozygous parent's transmissions, relative to the same region in homozygous parents — this is directly testable in existing long-read pedigrees and controls for sequence composition, because the same sequence is compared. (ii) The Yq12 rate (1.99×10^{-7} ⁵) should be explained by the same mechanism rather than by satellite composition alone, and should therefore be reproduced in autosomal satellite blocks that *do* synapse — if HSat3 on 1q, 9q or 16q shows a *lower* DNM rate than acrocentric HSat3, composition is not the driver and synapsis is. This is the cleanest single experiment we can propose from this report, and it requires no new sequencing beyond existing pedigree data re-analysed with a complete reference.

21.2 The acrocentric short arms are five autosomal non-recombining haplotype systems

If allelic crossover is genuinely absent across the p-arm, each short arm is inherited as a single linkage unit — an autosomal analogue of the male-specific Y or the mitochondrial genome, but present in two copies in every individual of both sexes, five times over. Combined with the tenfold mutation rate, each arm accumulates roughly 10× more substitutions per generation than euchromatin with no recombination to erase the genealogy.

Consequences, if true. Short arms should carry deep, well-resolved, population-structured haplogroup phylogenies; they should be exceptionally informative for recent demography and relatedness; ectopic exchange should appear as discrete reticulation events in those phylogenies rather than as a continuous recombination background; and hybrid arms should be datable.

Predictions. (i) Linkage disequilibrium measured across a short arm (using Regime I/II variants only) should approach unity within an arm and drop discontinuously at the centromere. (ii) A phylogeny built from Regime I short-arm variants across diverse individuals should be tree-like with a small number of clear reticulations, not net-like. (iii) The five arms should give five partially concordant, partially independent genealogies within the same individual. All three are testable on existing HPRC and HGSC assemblies restricted to the arm fraction that does assemble — which is a tractable analysis today, and to our knowledge has not been done.

21.3 rDNA dosage as a hidden modifier in rare disease

rDNA copy number is heritable, varies two- to three-fold between individuals, is coupled to gene expression and mitochondrial abundance^{20–21}, and sits upstream of a pathway whose disruption causes tissue-specific developmental disease⁵¹. It is also measurable from short-read data that pediatric rare-disease cohorts already hold.

Prediction. rDNA copy number should behave as a modifier of penetrance or severity for phenotypes with a nucleolar or ribosomal component — craniofacial, haematopoietic and growth phenotypes in particular — and as a modifier of the phenotypic consequences of aneuploidy. **This is testable at zero sequencing cost** in any existing cohort with short-read WGS and deep phenotyping: estimate rDNA copy number per individual, test for association with severity within diagnosed groups, and test for a copy-number shift in undiagnosed versus diagnosed cases. We consider this the single most actionable idea in this report for a group holding an existing pediatric WGS cohort.

21.4 Interlocus gene-conversion tomography between arms

Fourteen of 103 short-arm de novo SNVs matched an allele present elsewhere in the same repeat family, consistent with interlocus gene conversion⁵. Conversion events are directional and pairwise, which means a pedigree provides a direct measurement of *conversion flux* between specific arms.

Prediction. Flux should be strongly asymmetric and should mirror the paralogy structure in Figure 5 — highest between 13p and 21p and between 14p and 22p, lowest involving 15p. Measuring this requires only that conversion-like DNMs be annotated by donor locus rather than discarded as ambiguous, across enough transmissions. It would yield the first direct map of sequence flow between human non-homologous chromosomes, and it provides an independent estimate of the ectopic exchange rate that is far better powered than counting whole-arm recombinants (one event in 13 transmissions).

21.5 Nucleolar function as a selective filter on arm haplotypes

If hybrid arms, rDNA morph composition and array lengths affect NOR competence, then arm haplotypes are not neutral, and their population frequencies should deviate from what mutation and drift alone predict.

Prediction. rDNA morph frequencies should show less variance between individuals than unit-level mutation and conversion rates predict — i.e. evidence of stabilising selection on morph composition — and individuals with extreme total rDNA copy number should show compensatory shifts in the active/silent ratio measurable by methylation. Both are testable with *ribotin* morph catalogues plus long-read methylation on a modest panel.

21.6 Methylation as an orthogonal, cheap haplotype tag

CDR position relative to the α -satellite array start, SST1 methylation level, and satellite-block boundary methylation are all haplotype-specific, stable enough to be transmitted, and obtained free with long-read data^{5,52}. Where sequence identity prevents haplotype discrimination, the epigenetic state may not.

Prediction. A methylation-derived fingerprint over the short arm should distinguish homologous arms within an individual at higher accuracy than sequence alone in Regime II/III, and should track transmission. If so, "epigenetic haplotype tags" become a practical tool for phasing satellite arrays — a use of methylation data that is currently not exploited at all.

22. Seven concrete proposals, and one of them built

These are specifications rather than speculations: each could be built with current data and tooling.

22.1 Anchor-relative coordinates for the short arms

Problem. Absolute coordinates do not transfer between haplotypes (20.1).

Proposal. Address short-arm positions relative to the nearest stable, single-copy-per-arm anchor, with an explicit compartment label and offset:

13p:DJ+142, 318 · 21p:SST1[3]+1, 204 · 15p:H0R(S2C15H1L)-88, 150

The anchor set is small and already defined: the subtelomeric ACRO repeat, the DJ, the rDNA array boundaries, the PJ, each named satellite array boundary, and the active HOR array start. Each anchor is identifiable by sequence in any assembly, so the address is computable without a reference. Within an array, the address is (array identifier, unit index, offset in unit) — which is stable against array length change in a way that base coordinates are not.

What it buys. Addresses that mean the same thing in two individuals whose arrays differ in length; clinical reports that remain valid when the reference is updated; a natural key for a database of array-level observations. **What it costs.** A translation layer, and a convention for choosing among multiple candidate anchors. We suggest this is a better use of community effort than any further liftover chain.

22.2 An acrocentric pangenome module with an explicit ambiguity model

Problem. Current human pangenome graphs inherit the assembly gaps and represent paralogy implicitly (20.5, 7.2).

Proposal. Build a dedicated acrocentric graph from the subset of assemblies with validated short arms, with three features current graphs lack: (i) nodes annotated with the *set* of arms they are compatible with, not a single position — an explicit representation of the 22–61% of each arm that has a $\geq 99\%$ twin elsewhere (Figure 5); (ii) array nodes typed as collapsible with a length distribution attached, rather than expanded to one arbitrary length; (iii) rDNA represented as a morph catalogue node rather than a sequence path.

We suggest thinking of the resulting structure as an **onion**: an outer layer of arm-specific sequence, a middle layer shared between one pair of arms, an inner layer shared among all five, and a core (rDNA, HOR arrays) that is not path-resolvable at all. Analysis should proceed layer by layer, reporting results at the outermost layer at which the data can resolve them — rather than forcing everything to the base level and silently mislabelling the inner layers.

22.3 An assignability score to replace MAPQ

Problem. MAPQ measures uniqueness against one reference haplotype and is misleading here (11.1, 20.4).

Proposal. A per-read posterior over arm-of-origin, computed from satellite "dialect" statistics. The pieces exist: satellite dialects — arm-characteristic variant and *k*-mer composition within a satellite family — were described in the original pangenome analysis of these regions⁴; complete assemblies provide training data; the inference is a standard mixture problem.

Concretely: (i) from validated short arms, build for each (arm $\times$ satellite family) a profile of discriminative *k*-mers or PSV allele frequencies; (ii) for each read, compute the likelihood under each profile; (iii) report a posterior over the five arms and a null "unassignable" class, fitted by expectation-maximisation over the whole read set so that the individual's own array-size mixture informs the prior. Output a posterior, not a binary placement.

This changes downstream practice: variant calls carry an arm-assignment posterior; the callable region becomes a probabilistic quantity; and a read that is genuinely ambiguous is recorded as ambiguous rather than placed. It is, we think, the most important missing piece of infrastructure for this region.

22.4 An array-level variant representation

Problem. VCF cannot represent Regime III variation (13.2, 20.7).

Proposal. A companion record type — call it an **array-level feature (ALF)** record — keyed by anchor-relative array identity rather than by base coordinate. Minimum fields:

Field	Content
array_id	anchor-relative identifier, e.g. 15p:HSat3[1]
sample, haplotype	sample and haplotype identifier
length, length_ci	estimated array length and interval
units, unit_spectrum	unit count and composition (e.g. HOR <i>n</i> -mer frequencies)
method, evidence	estimator, input data type, depth
flags	collapsed, unresolved, boundary-ambiguous, model-derived
methylation	mean and profile summary where available
assignability	posterior that the array is correctly assigned to this arm (22.3)

Two design decisions matter. First, **length is a measurement with an uncertainty**, not a genotype — the format must carry the interval, and downstream statistics must use it. Second, **the flags are mandatory**, because the commonest error in this field is treating a collapsed array as a short one. An ALF record set is also the natural input to array-trait association testing (20.8).

22.5 Per-chromosome rDNA copy number by IGS-barcode deconvolution

Problem. Total rDNA copy number is measurable; per-chromosome copy number is not (section 16).

Proposal. The intergenic spacer carries far more variation than the transcribed unit, and rDNA arrays are homogenised *within* a chromosome more strongly than between chromosomes^{17–18}. Chromosome-characteristic IGS haplotypes should therefore exist. The proposal is a two-stage method: (i) on a long-read reference panel, use ribotin morphs together with DJ/PJ-anchored reads to establish which IGS morphs occur on which arm, building an arm $\times$ morph matrix; (ii) in a short-read sample, count morph-discriminative *k*-mers and solve the constrained non-negative deconvolution for per-arm copy number,

with total copy number as a constraint.

This is a tractable, well-posed inverse problem, and it would turn per-chromosome rDNA dosage — currently a FISH-only measurement — into something obtainable from existing cohort WGS. The main risk is that arm-specific morphs are not sufficiently distinct in all individuals; that risk is itself worth measuring, and stage (i) measures it.

22.6 A sequencing-based Robertsonian carrier test, and an acrocentric liability score

Problem. ROB carriers are found by karyotype and missed by sequencing (6.1, 20.10).

Proposal, part one — detection. A ROB carrier has lost most of two short arms. Three independent short-read signatures follow: reduced depth over arm-specific satellite *k*-mers for the two involved arms; a reduced total rDNA copy number (two arrays lost); and an altered ratio of arm-specific centromeric HOR *k*-mers. A classifier on these features, trained on known carriers — which cytogenetics laboratories hold in quantity — would give a validated, essentially free screen applicable retrospectively to every short-read cohort in existence. Given a population carrier frequency near 1 in 800, any cohort of 10,000 contains of order a dozen carriers, currently unrecognised.

Proposal, part two — risk stratification. Since ROB formation is NAHR between high-identity substrates shared between arms^{33,42}, an individual's risk should depend on their own haplotypes' shared-substrate content. Define an **acrocentric liability score** for each arm pair as a sum over shared blocks of (block length) × (identity), computed from that individual's assemblies — essentially a personalised version of Figure 5. The prediction is that the score is elevated in ROB carriers' transmitting parents and in families with recurrent ROBs, and that the population distribution of the score predicts the observed 13;14 > 14;21 > 13;21 hierarchy. This is testable on existing ROB cohorts with long-read sequencing of parents, and it would be the first mechanistic risk model for the commonest human structural rearrangement.

22.7 An acrocentric benchmark

Problem. No truth set exists (20.2).

Proposal. Build one on the CEPH1463 pedigree, for which validated short-arm assemblies, three data types and transmission-based validation already exist^{5,106}. Components:

1. **Confident regions**, defined per haplotype by the callable-region intersection in 14.1, published as anchor-relative intervals and as coordinates on each assembly.
2. **Truth variants** in Regimes I and II, validated by orthogonal technology and by transmission.
3. **Truth array-level features** (ALF records, 22.4) for every array that assembles cleanly, with lengths and unit spectra.
4. **Stratifications** that matter here: by regime; by satellite family; by cross-arm paralogy level (using the ≥90/95/99% strata of Figure 5); by distance from the nearest single-copy anchor; and distal versus proximal.
5. **Assignability-aware metrics.** Conventional precision and recall are insufficient: a call that is correct in sequence but assigned to the wrong arm should be scored as a distinct error class, because that is the dominant error mode. We propose reporting *arm-assignment accuracy* alongside genotype concordance, and reporting both stratified by paralogy level.

Even components 1 and 4 alone, published without truth variants, would let the field compare pipelines for the first time.

22.8 Implemented: a learned slot grammar, and the placement of unlabelled sequence

Sections 22.1, 22.3 and 22.4 proposed anchor-relative coordinates, an assignability score and an array-level representation. This section reports a working implementation of the first and third, an evaluation of the second, and one instructive failure. Code and tables accompany this report; methods in section 25.

The dataset

Per-haplotype satellite annotations were assembled for **944 acrocentric short arms**: 789 from 94 unrelated individuals of HPRC release 2, annotated by the CenSat pipeline, and 155 from the CEPH1463 pedigree, annotated by the targeted track that additionally labels ACRO, the DJ, the PJ and SST1. Figure 9 draws all of them.

944 acrocentric short arms: 789 from 94 unrelated HPRC individuals, 155 from one pedigree
one row per haplotype, sorted by length within each chromosome

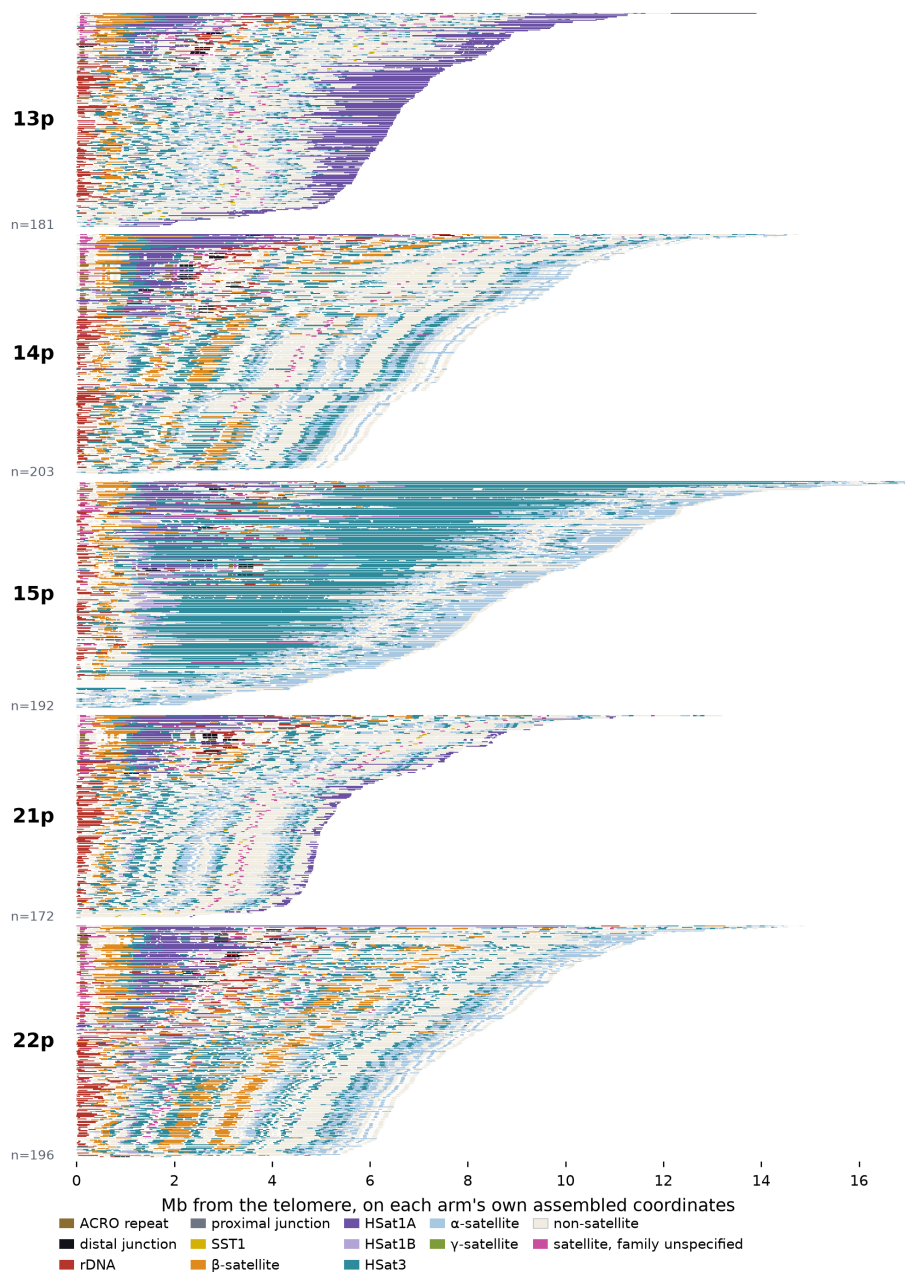


Figure 9. Nine hundred and forty-four acrocentric short arms. One row per haplotype, grouped by chromosome and sorted by length, drawn on each arm's own assembled coordinates. Arms begin at the start of assembled short-arm sequence, which for most HPRC haplotypes is the rDNA array itself because distal sequence is rarely assembled — hence the red left margin. The banding within each chromosome is the conserved block order of section 5.5; the diagonal wedges are size variation in a single array, most visibly the proximal HSat1A block of 13p and 21p and the HSat3 block of 15p. Magenta marks satellite that the annotation calls satellite without naming a family: 120 Mb of it, in 56% of arms.

A grammar with slots

Because the block order is conserved, an arm can be written as a chain of **slots**, each carrying a distribution over block families, with alignment allowing a slot to be skipped and a block to be inserted between slots — the structure of a profile hidden Markov model, with satellite families in place of residues. Two design choices make it work here. Arms are split at the rDNA array, the one landmark present on every arm, and the distal and proximal segments are modelled separately, since the distal grammar is shared across chromosomes and the proximal grammar is not. And the symbol for *satellite of unspecified family*

emits with probability one from every slot, so unlabelled blocks are positioned by their context alone and contribute nothing to the slot distributions they are then compared against.

Slot chains are seeded from an observed arm — the medoid of the group by n-gram similarity — and refined by Viterbi EM, with slots seeded on a single-copy landmark held fixed so that the anchors cannot be re-purposed. Fitted models run from 10 slots (15p proximal) to 21 (14p proximal). Figure 10 shows every arm mapped into the resulting coordinates.

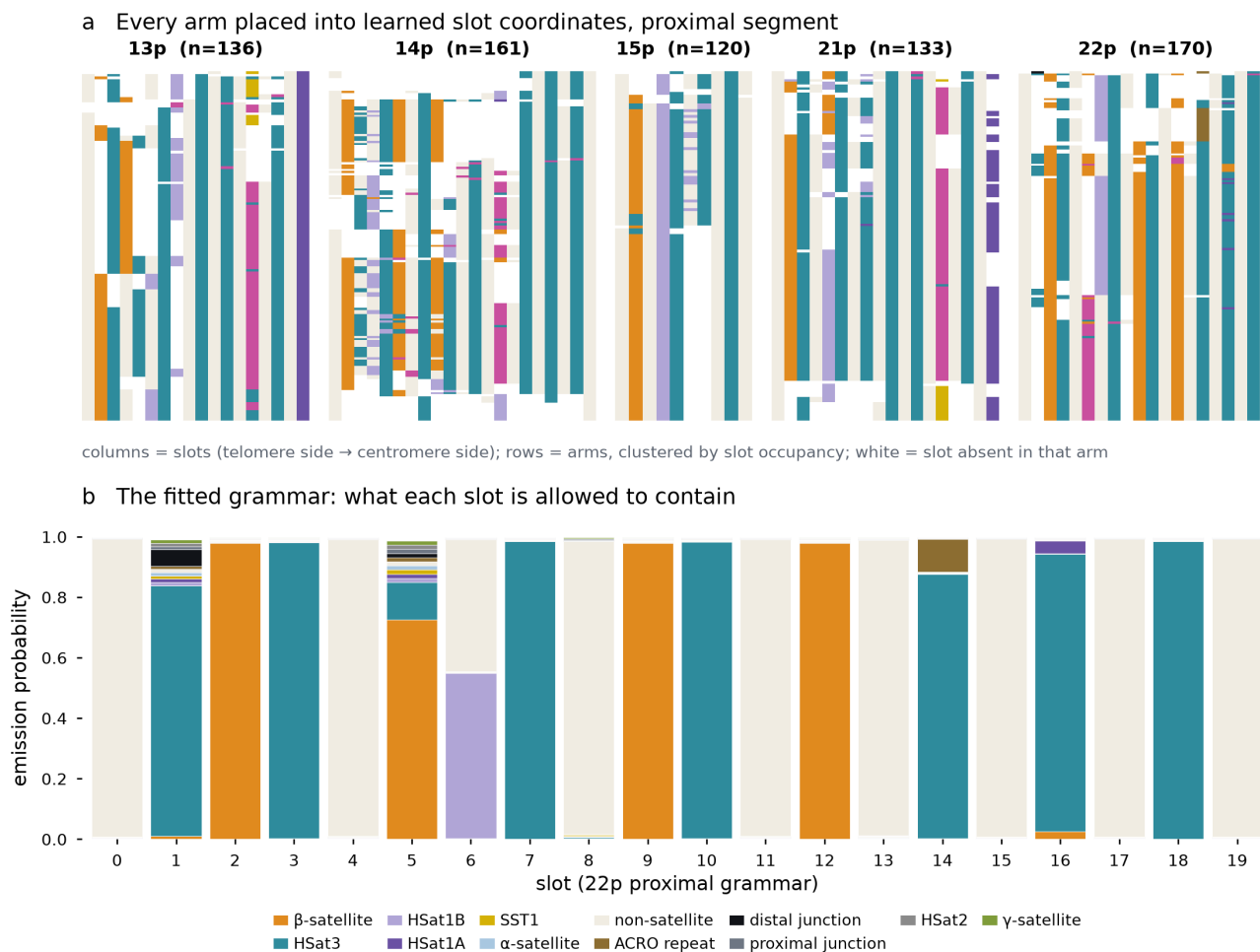


Figure 10. The grammar, and every arm expressed in it. (a) Each row is an arm, each column a slot of that chromosome's fitted proximal grammar, coloured by the family assigned by Viterbi alignment; white means the slot is absent from that arm. Rows are clustered by slot occupancy. Solid columns are slots whose content is effectively fixed across the population; magenta columns are slots consistently occupied by satellite the annotation does not name. (b) The fitted emission distribution of each slot of the 22p proximal grammar — what the grammar permits at each position.

Recovering a block's identity from its position

The natural test of a grammar that claims to know what belongs where is to hide a block and ask it. Masking one block at a time in arms from **held-out individuals**, the grammar names 89% of masked blocks and is correct for 92.1% of those — 82.3% of all masked blocks, against 69.7% for a position-bin baseline, 64.1% for majority class and 35.0% for copying the nearest labelled neighbour (Figure 11a). Trained on HPRC and tested on the pedigree it holds up: 80.5% overall. The posterior is usable as a confidence: above 0.95 accuracy is 95–99% (Figure 11b).

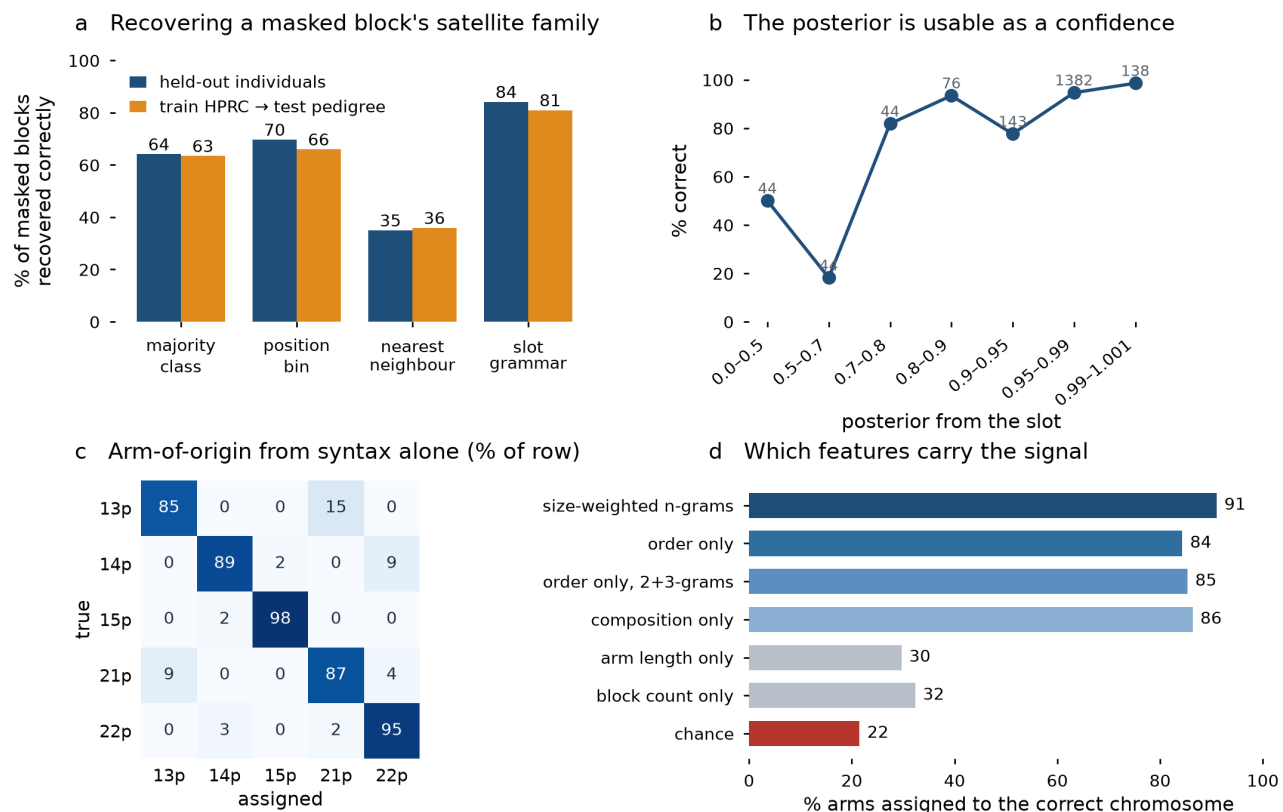


Figure 11. Evaluation. (a) Recovery of a masked block's satellite family, for held-out individuals and across datasets, against three baselines. (b) Accuracy by posterior bin; the annotation above each point is the number of blocks in that bin. (c) Assignment of an arm to its chromosome from satellite syntax alone, as a percentage of each true class. (d) Which features carry that signal.

Arm-of-origin from syntax alone

Representing each arm as size-weighted counts of ordered block n-grams and classifying by nearest neighbours, **91.0% of arms from held-out individuals are assigned to the correct chromosome**, against 21.5% for the largest class (Figure 11c–d). Arm length alone gives 29.7% and block count 32.3%, so the signal is in composition and order, not size; order alone, with block sizes discarded, still gives 85.3%.

The errors are the interesting part. They fall almost entirely on 13p–21p (15% of 13p arms assigned to 21p, 9% the reverse) and 14p–22p (9% and 3%) — the two pairs that share an active centromeric HOR family, carry the highest cross-arm paralogy (Figure 5) and account for the recurrent Robertsonian translocations. A classifier built only on satellite block syntax rediscovers the pairing structure of the acrocentric community without being told about it.

Placing the unlabelled 120 Mb

Applied to the 735 blocks that carry no family assignment, the grammar gives 92% of them a **reproducible slot address** — a position in the canonical arm, which is the anchor-relative coordinate of section 22.1 obtained by fitting rather than by decree — and names 88% of them. The results divide into three honest categories.

Validated. 319 blocks totalling 55.1 Mb, median 147 kb, are called ACRO repeat. This is a vocabulary gap rather than a discovery: the CenSat annotation has no ACRO label, so genuine subtelomeric ACRO sequence in HPRC assemblies necessarily lands in the unspecified bin. Masking *labelled* ACRO blocks in pedigree arms recovers them at 85.5% (n=119), which is the evidence that these calls are real.

Plausible but not validated. 98 blocks on 21p, median 71 kb, are called SST1 — squarely inside the published 21–104 kb range for SST1 arrays, at a slot the pedigree annotation labels SST1, on a chromosome known to carry it. But the corresponding masking test fails: masked pedigree SST1 blocks are recovered at 0% because pedigree and HPRC arms take different Viterbi paths through the grammar. The call is well-motivated and independently consistent in size, and it is not

demonstrated. It would be settled in an afternoon by aligning the SST1 consensus to those coordinates.

Recurrent and unnamed. Six slots hold unspecified satellite in at least a quarter of their chromosome's arms (data/unknown_slot_summary.csv). The most prominent is a proximal slot on 13p present in 62% of arms, 8.4 Mb in total, median block size 85 kb, for which the grammar offers no confident family (median posterior 0.49); 14p has a counterpart at 38.5% of arms and median 75 kb. Given that SST1 is documented on 13p, 14p and 21p, and that the equivalent 21p slot is confidently called SST1 with a matching size distribution, the parsimonious reading is that these are the SST1 arrays of 13p and 14p, unnamed because the annotation vocabulary omits them. The alternative — that they are a distinct, positionally fixed, unnamed satellite — is the more interesting possibility and is equally testable.

What this does and does not deliver

It delivers a fitted, inspectable description of where each block type belongs on each acrocentric short arm; a coordinate for any block, including unlabelled ones, that is defined by grammatical position rather than by base offset in one assembly; a calibrated family prediction for unlabelled satellite where the grammar has evidence; and a chromosome assignment from syntax alone whose errors are biologically interpretable.

It does not deliver a sequence-level method. Everything here operates on annotation tracks, so it inherits their errors and their vocabulary, and it cannot discover a satellite family that no annotation names. The obvious next step is to fit the same grammar to blocks defined directly from sequence — by *k*-mer composition of each interval rather than by a label — which would make the "recurrent and unnamed" category resolvable instead of merely catalogued.

The failure worth reporting

The representation does not transfer between annotation pipelines. Against 91.0% within a single pipeline, chromosome assignment drops to 40.8% training on the pedigree track and testing on HPRC's CenSat track — still above the 21.5% largest-class rate but far below usable — and to 23.2%, indistinguishable from chance, in the reverse direction. Masked-block recovery degrades in the same direction (33.8% training on the pedigree, against 80.5% training on HPRC). The cause is granularity: the two pipelines cut the same sequence into different numbers of blocks, so *n*-gram profiles and Viterbi paths are not comparable across them, and the pedigree's 155 related arms are in any case thin training data.

This is a concrete, measured obstacle to the assignability score proposed in section 22.3, and it sharpens that proposal. A portable arm-assignment tool cannot be built on annotation output as currently distributed; it needs either a harmonised satellite annotation applied uniformly to every assembly, or features computed from sequence rather than from labels. Of the two, the second is more likely to work, and it is the direction we would now recommend.

23. Cookbook

23.1 If you have short reads only

1. Align to T2T-CHM13v2.0 if starting fresh; otherwise keep GRCh38 and add explicit short-arm exclusions.
2. Carry the accessibility mask; report the 2.82 Mb accessible denominator.
3. Report short-arm SNV/indel calls only within the mask; state that the remainder was not assessed.
4. Compute total rDNA copy number by *k*-mer counting against the KY962518 unit, normalised within cohort.
5. Compute relative satellite array content per family (HSat1A, HSat1B, HSat2/3, β -satellite, SST1, α -satellite HOR by family) by *k*-mer depth.
6. Flag outliers on 4 and 5 for long-read follow-up.
7. Never report absence of variation.

23.2 If you have HiFi only (~30×)

1. Assemble with hifiasm (UL mode if any ONT is available) or Verkko2.
2. Expect proximal short-arm sequence to assemble, rDNA to collapse, distal sequence to be orphaned or missing.
3. Anchor contigs through ≥ 1 Mb of q-arm alignment at least 1 Mb from the centromere.

4. Run NucFreq and Flagger; exclude the union.
5. Annotate satellites, SDs, DJ/PJ, rDNA, telomere.
6. Run `ribotin` for rDNA morphs.
7. Report array lengths with collapse flags, per haplotype.

23.3 If you have HiFi + UL-ONT + Hi-C

1. Assemble with Verkko2 using all three data types.
2. Identify pq-scatis; expect ~97% of haplotypes recovered, ~41% with distal sequence scaffolded.
3. Validate with NucFreq, Flagger, ModDotPlot self-dotplots and CenMAP for centromeres.
4. Annotate as in 23.2; add CDR identification and methylation.
5. Produce the callable-region set (14.1) and report it.
6. Compare haplotypes all-vs-all with `minimap2 -x asm20 -c --eqx -D -P --dual=no` and 10 kb sliding-window identity.

23.4 If you have a trio or pedigree

Follow 23.3 for every member, then the de novo procedure in 14.2. Non-negotiable steps: personalised parental reference; two-technology orthogonal support; the 10 kb sliding-window identity filter against distal/proximal misalignment; manual review; gene-conversion annotation of every candidate. Use any further generation as transmission validation.

23.5 If you have a Robertsonian or other rearrangement carrier

1. Verkko2 with Hi-C; the graph, not the alignment, is the primary evidence.
2. Expect a fused contig carrying two long arms; characterise the junction by PSV analysis against both parental or reference arms.
3. Confirm with karyotype and, if available, Strand-seq.
4. Report the breakpoint anchor-relative to SST1 and to the DJ/PJ, not only as reference coordinates.

24. Reporting standards for acrocentric analyses

We suggest that any paper making a claim about the acrocentric short arms should report, in methods or a supplementary table, the following minimum set. Most current papers report none of it.

1. Reference assembly *and* version, and annotation versions (CenSat, RepeatMasker, SD call set).
2. Whether rDNA models were masked, and by which intervals.
3. The callable denominator, per sample, per haplotype, per chromosome — and how it was computed.
4. The accessibility or mappability criterion applied, with the track cited.
5. Assembly QC outcome: QV, NucFreq/Flagger-flagged fraction, whether distal sequence was recovered and how it was verified (ACRO repeat, DJ).
6. For array lengths: estimator, collapse-detection method, and uncertainty.
7. For rDNA: unit reference used, masking, estimator, and whether the estimate is total or per-chromosome.
8. Arm-assignment procedure, including the anchoring requirement, and the count of contigs that failed it.
9. For variant calls: the orthogonal support requirement, and the gene-conversion check.
10. Explicit statement of what was *not* assessed.

A single sentence — "the acrocentric short arms were not assessed; *n* Mb excluded" — is a complete and honest fulfilment of this standard for a study that excludes them, and is far preferable to silence.

25. Methods for the analyses in this report

All quantitative statements about reference content were computed for this report from public T2T-CHM13v2.0 resources. No new sequencing data were generated or used.

Region definitions. Each acrocentric short arm was defined as the interval from position 0 to the start of the active α -satellite higher-order-repeat array, identified as the largest suprachromosomal-family-2 "live" HOR run in the CenSat v2.1 annotation: hor_13_3 (S2C13/21H1L) at 15,548 kb; hor_14_3 (S2C14/22H1L) at 10,092 kb; hor_15_4 (S2C15H1L) at 16,679 kb; hor_21_3 (S2C13/21H1L) at 10,963 kb; hor_22_9 (S2C14/22H1L) at 12,788 kb. GRCh38 comparisons use the cytogenetic p-arm (last p-band end in the UCSC cytoBand table) and the UCSC gap table for N intervals.

Composition. Satellite family composition was computed by merging CenSat v2.1 intervals by family within each arm and summing clipped lengths. Family labels follow the CenSat vocabulary; ct intervals (non-satellite "transition" sequence within the annotated region) are reported as non-satellite.

Segmental duplications. Cross-arm paralogy was computed from the SEDEF call set distributed for CHM13v2.0 (chm13v2.0_SD.full.bed), retaining pairs in which both mates fall inside a short arm as defined above. Identity was taken as matchB/aInB. Per-arm coverage was computed by merging query intervals at each identity threshold and clipping to the arm. Note that this call set applies the conventional filter excluding duplications more than 70% satellite, so these figures are a lower bound on total paralogy; the median satellite fraction of the retained acrocentric–acrocentric pairs is 0.26.

Short-read accessibility. The T2T-CHM13v2.0 short-read accessibility mask (combined_mask.bed) was intersected with arm and compartment intervals. Polarity was confirmed by reproducing the per-chromosome accessible-base counts in the distributed reference_accessibility_comparison.txt table. The autosomal reference figure of 86.9% is the sum of accessible bases over the sum of total bases for chr1–chr22 from that table.

Unique-anchor analysis. The minimum-unique- k -mer-length track chm13v2.0.mul.bw gives, for each position, the length of the shortest substring starting there that is unique in the genome. Per-arm ladders (Figure 3a) report the fraction of positions with a value $\leq L$ for a grid of L . The euchromatic reference is chr13:40,000,000–60,000,000. Difficulty profiles (Figure 3b) and regime maps (Figure 6) use the median value in 10 kb windows. Regime boundaries are 1 kb and 100 kb.

rDNA. Array extents are the CenSat rDNA intervals. Modelled unit counts are array length divided by 44,838 bp. The unit feature map (Figure 4a) was parsed from the GenBank record KY962518.1.

Block order across haplotypes (sections 5.5, Figures 7 and 8a). Per-haplotype satellite annotations for the 161 CEPH1463 acrocentric arms were taken from the all_samples_censat.bed track of the Platinum Pedigree AcroMutRecomb repository⁵; independent annotations for HPRC release 2 came from the per-haplotype censat.bed files under working/HPRC/<sample>/assemblies/release2/annotation/censat/, with scaffold lengths from the accompanying .fai and centromere intervals from active.centromeres.bed. HPRC scaffolds were assigned to an acrocentric chromosome by the suprachromosomal-family-2 active HOR family named in the annotation (S2C13/21H1L, S2C14/22H1L, S2C15H1L), disambiguated within a family by scaffold length against the corresponding CHM13 chromosome length, retaining only chromosome-scale scaffolds (within 15% of the expected length) and one scaffold per haplotype and chromosome. Scaffolds whose centromere fell in the distal half were reverse-complemented in coordinate space. The short arm was taken as the interval from the telomere-proximal end to the start of the active array. Annotation intervals were merged into blocks by family, joining intervals of the same family within 50 kb and discarding blocks below 50 kb, and DJ-arm and DJ-flank labels were merged. Order consistency for a pair of families is the fraction of arms in which one family's largest block lies distal to the other's, taken as $\max(f, 1-f)$ over arms containing both families, requiring at least six informative arms per pair; reported segment-level values are weighted by the number of informative arms. Analysis was stratified into distal and proximal segments relative to the rDNA array because HSat1A and HSat3 occur on both sides of it.

Per-compartment mutation rates (section 5.4, Figure 8b–c). De novo SNV counts per satellite family are those reported by Lin et al.; denominators were computed here. Callable bases per compartment were obtained from the same per-haplotype annotation over the interval from the telomere to the end of the active α -satellite array — the scope of the published mutation analysis — after subtracting the union of Flagger- and NucFreq-flagged intervals distributed with that resource, and excluding gap and telomere labels. The study's callable denominator, 26 haplotype-transmissions $\times$ 30.7 Mb, was apportioned across compartments in the resulting proportions; this reproduces the published overall rate (1.29×10^{-7} against 1.33×10^{-7} reported), which is the check that the apportionment is approximately right. The 12 mutations reported in segmental duplications that are

not satellite were assigned to the non-satellite compartment together with the 21 falling outside both annotations. Intervals on per-compartment rates are 95% normal intervals on the Poisson count; the satellite/non-satellite contrast is a binomial test of the observed split against the split expected from callable bases.

Satellite syntax, the slot grammar and placement (section 22.8, Figures 9–11). Arms were assembled from two annotation dialects mapped onto one alphabet of satellite families, landmarks, a non-satellite spacer class and a wildcard for satellite whose family the annotation leaves unspecified. Intervals were merged into blocks by family, joining intervals of the same family within 50 kb and discarding blocks below 50 kb. HPRC release 2 arms were taken from per-haplotype CenSat annotations; scaffolds were assigned to an acrocentric chromosome by the suprachromosomal-family-2 active HOR family named in the annotation, disambiguated within a family by scaffold length against the corresponding CHM13 chromosome, restricted to chromosome-scale scaffolds within 15% of the expected length, one per haplotype and chromosome, and reverse-complemented in coordinate space where the centromere fell in the distal half. This yielded 789 arms from 94 individuals; the remaining release 2 haplotypes were excluded because their acrocentrics are not assembled to chromosome scale. Arms were split at the rDNA array and the distal and proximal segments modelled separately. Each model is a left-to-right profile whose slots carry a distribution over families, with fixed-cost slot deletion and block insertion; slot chains were seeded from the group medoid by n-gram cosine similarity and refined by Viterbi EM for up to twelve rounds, with slots seeded on a single-copy landmark held at their seeded distribution. The wildcard emits with probability one from every slot and contributes no emission counts. A family prediction is the slot distribution renormalised over satellite families, reported only when the slot carries at least 0.25 total probability on that alphabet — without that gate a slot dominated by non-satellite sequence still returns its highest-probability satellite family from a vanishing probability. Masked-block recovery masks one block at a time, re-aligns, and compares the prediction with the held-out truth; baselines are the majority family, the modal family of the block's decile of segment position, and the family of the nearest labelled block. Splits are by individual, so no individual contributes to both training and test. Chromosome assignment uses size-weighted counts of ordered block n-grams (orders 1–3, weights the square root of mean block size, L2-normalised) with five-nearest-neighbour classification under cosine distance, and the reported controls vary the weighting and the n-gram orders. Random seed 7 throughout.

Reproducibility caveat. Two of these measurements are properties of the reference rather than of human genomes, and are labelled as such throughout: any statistic computed inside the CHM13 rDNA models, and any mappability or accessibility figure, which describes the CHM13 haplotype and not the population.

Software. Analyses used Python 3 with NumPy, pandas and Matplotlib, and pyBigWig for track access. Bibliographic metadata were retrieved programmatically and each cited reference verified by title match before inclusion.

26. Glossary

Term	Definition
ACRO repeat	satellite family at the subtelomeric distal end of acrocentric short arms; the marker for distal assembly completeness
ALF record	array-level feature record; the representation proposed in 22.4
α-satellite (aSat)	171 bp-monomer satellite; organised into higher-order repeats (HOR) at centromeres, monomeric in pericentromeric transitions
Assignability	probability that a read is placed on the correct arm and haplotype in the sample's own genome; distinct from mappability
β-satellite	GC-rich ~68 bp-monomer satellite characteristic of acrocentric short arms
CDR	centromere dip region; hypomethylated subdomain of the active HOR array marking kinetochore attachment
CenSat	the T2T-CHM13 satellite annotation (v2.1 used here)
Distal / proximal	positions relative to the rDNA array: distal is telomere-side, proximal is centromere-side
DJ	distal junction; ~300 kb conserved segmental duplication immediately distal to the rDNA array, present on all five arms
HOR	higher-order repeat; a multi-monomer α -satellite unit, tandemly repeated
HSat1A/1B, HSat2, HSat3	human satellite families; HSat1A is AT-rich with a ~42 bp repeat, HSat3 forms classical C-band heterochromatin
IGS	intergenic spacer; the 31.5 kb non-transcribed portion of the rDNA unit
Mappability	uniqueness of a substring in a given reference assembly
Morph	a recurrent rDNA unit haplotype; the natural allele unit for rDNA

Term	Definition
MU track	minimum-unique- <i>k</i> -mer-length track; per-base shortest genome-unique substring length
NAHR	non-allelic homologous recombination
NOR	nucleolus organizer region; an rDNA array plus junctions, functionally competent to nucleate a nucleolus
PHR	pseudo-homologous region; high-identity segment shared between non-homologous acrocentric arms, implicated in ectopic exchange
PJ	proximal junction; the segmental duplication flanking rDNA on the centromeric side
pq-scatig	scaffolded contig spanning short arm, centromere and ≥ 1 Mb of uniquely-alignable long arm
PSV	paralog-specific variant; a position distinguishing two paralogous copies
rDNA	ribosomal DNA; the 45S/47S rRNA gene array (5S rRNA is elsewhere, at 1q42)
Regime I/II/III	analysis regimes defined by minimum-unique-anchor length (≤ 1 kb, 1–100 kb, >100 kb)
ROB	Robertsonian translocation; whole-arm fusion of two acrocentric chromosomes
SAAC	short arm of an acrocentric chromosome
SD	segmental duplication
SST1 / NBL2	~ 1.4 kb-unit macrosatellite on 13p, 14p and 21p; marks the PHRs and is implicated in ROB formation

27. Resources

Reference and annotation (T2T-CHM13v2.0). Available from the Human Pangenomics S3 bucket `s3-us-west-2.amazonaws.com/human-pangenomics/T2T/CHM13/assemblies/annotation/` and mirrored at UCSC as assembly `hs1`:

- `chm13v2.0_censat_v2.1.bed` — satellite annotation
- `chm13v2.0_SD.bed`, `chm13v2.0_SD.full.bed` — segmental duplications, with identity in the full file
- `accessibility/combined_mask.bed.gz` and component `mapq_`, `coverage_`, `baseq_` masks — short-read accessibility, with GRCh37/GRCh38 equivalents for comparison
- `mappability/chm13v2.0.mul.bw`, `.mur.bw` — minimum-unique-*k*-mer-length tracks
- `chm13v1.1.rdna_model.bed`, `chm13v1.1.rdna_units.bed` — rDNA model intervals (mask these)
- `chm13v2.0_RepeatMasker_4.1.2p1.bed`, `chm13v2.0_cytobands_allchrs.bed`, `chm13v2.0_telomere.bed`
- `regulation/`, `nonB/`, `pattern/` — methylation, non-B DNA and microsatellite tracks

Sequence standards. GenBank KY962518.1 — human rDNA complete repeating unit, 44,838 bp. Rotate to begin upstream of the 45S transcription start when using as an alignment target. HSat consensus sequences from the HSat review repository²⁸.

Assemblies and pedigree data. HPRC and HGSC assembly releases^{2–3}; the CEPH1463 / Platinum Pedigree short-arm assemblies and annotations^{5,106}, with code and the per-haplotype satellite annotation track at the AcroMutRecomb repository.

Per-haplotype satellite annotations, useful and easy to miss. Two resources make the analyses in sections 5.4 and 5.5 reproducible without assembling anything. The Platinum Pedigree repository publishes `annotation/all_samples_censat.bed` (satellite blocks for all 161 pedigree arms, including DJ, PJ, ACRO and SST1 labels that the standard CenSat vocabulary lacks) together with `annotation/flagger_nucfreq_merged.bed` (the assembly-error exclusion set). HPRC release 2 publishes per-haplotype CenSat annotations at `s3://human-pangenomics/working/HPRC/<sample>/assemblies/release2/annotation/censat/<sample>_hap<N>_hprc_r2_v1.cenSat.bed`, with `active.centromeres.bed` and the assembly `.fai` alongside. Both are small BED files; neither requires downloading an assembly.

Software. Verkko2⁷⁴; hifiasm^{84–85}; ribotin²⁷; Merqury⁸⁸; NucFreq/NucFlag and Flagger^{2,7}; ModDotPlot⁸⁹; CenMAP and HumAS-HMMER³⁵; SEDEF⁹⁵; minimap2⁸⁷; wfmash and PGGB⁷⁰; Minigraph-Cactus⁷¹; odgi⁹²; SVbyEye⁹⁰; wgatools⁹⁶; sniffles2¹⁰⁰; Truvari¹⁰⁷; TRGT⁹⁸; DeepPolisher⁸⁶; compleasm¹⁰⁸; BEDTools¹⁰⁹.

28. A closing assessment

The acrocentric short arms have moved in four years from unsequenced to sequenced, and from unmeasured to partly measured. What has not happened is the corresponding change in analytical practice: the regions remain excluded from essentially every cohort analysis, and the exclusions are usually silent.

Our reading of the numbers assembled here is that the field is at an inflection point with an unusually favourable cost structure. The hardest single obstacle — the rDNA array — accounts for 9.93 Mb, and will need new technology. But it is the *only* compartment that needs new technology: every other satellite family is more than 97% anchorable with reads that many laboratories already generate. The remaining obstacles are infrastructural rather than physical: a coordinate system, an assignability metric, a data model for arrays, and a benchmark. None requires an instrument that does not exist. All four could be built, by a small number of groups, within a couple of years.

One structural finding makes that infrastructural programme easier than it looked. The block order in these arms is conserved — an invariant landmark backbone, a distal satellite order shared by all five chromosomes, and free ordering confined to a few identifiable slots (section 5.5). Almost all of the interindividual difference is length variation within that fixed syntax. A coordinate system and a data model only have to name slots and measure their sizes, which is a far more tractable specification than one that would have to accommodate arbitrary rearrangement.

Meanwhile there is work available at zero marginal sequencing cost. Total rDNA copy number, relative satellite array content and a Robertsonian carrier screen are all computable from short-read data that thousands of cohorts already hold; the rDNA-dosage-as-modifier hypothesis (21.3) and the synopsis-versus-composition test of mutation rate (21.1) are both answerable by re-analysis rather than by new experiments. For a group already holding a trio-based pediatric cohort, the most immediately valuable action is probably the least glamorous: compute the short-arm denominator that has always been implicit, state it, and then look at what falls inside it.

29. References

References are numbered in order of first appearance. Every entry was verified against bibliographic metadata retrieved programmatically; data resources are cited by file name and repository.

1. Nurk S, Koren S, et al.. *The complete sequence of a human genome*. Science **2022**. doi:10.1126/science.abj6987
2. Liao W, Asri M, et al.. *A draft human pangenome reference*. Nature **2023**. doi:10.1038/s41586-023-05896-x
3. Logsdon GA, Ebert P, et al.. *Complex genetic variation in nearly complete human genomes*. Nature **2025**. doi:10.1038/s41586-025-09140-6
4. Guarracino A, Buonaiuto S, et al.. *Recombination between heterologous human acrocentric chromosomes*. Nature **2023**. doi:10.1038/s41586-023-05976-y
5. Lin J, Mastroianni FK, Noyes MD, et al.. *Human acrocentric chromosome short-arm de novo mutation and recombination*. Cell **2026**. doi:10.1016/j.cell.2026.05.035
6. Human Pangenomics Consortium. *T2T-CHM13v2.0 CenSat v2.1 satellite annotation*. data resource **2023**. chm13v2.0_censat_v2.1.bed
7. Vollger MR, Guitart X, et al.. *Segmental duplications and their variation in a complete human genome*. Science **2022**. doi:10.1126/science.abj6965
8. Human Pangenomics Consortium. *T2T-CHM13v2.0 short-read accessibility mask*. data resource **2022**. chm13v2.0_combined_mask.bed
9. Human Pangenomics Consortium. *T2T-CHM13v2.0 minimum-unique-k-mer-length (MU) tracks*. data resource **2022**. chm13v2.0.mul.bw / .mur.bw
10. Floutsakou I, Agrawal S, et al.. *The shared genomic architecture of human nucleolar organizer regions*. Genome Research **2013**. doi:10.1101/gr.157941.113
11. McStay B. *The p-Arms of Human Acrocentric Chromosomes Play by a Different Set of Rules*. Annual Review of Genomics and Human Genetics **2023**. doi:10.1146/annurev-genom-101122-081642

12. Antonarakis SE. *Short arms of human acrocentric chromosomes and the completion of the human genome sequence.* Genome Research **2022**. doi:10.1101/gr.275350.121
13. McStay B. *Nucleolar organizer regions: genomic ‘dark matter’ requiring illumination.* Genes & Development **2016**. doi:10.1101/gad.283838.116
14. van Sluis M, Gailín MÓ, McCarter JGW, et al.. *Human NORs, comprising rDNA arrays and functionally conserved distal elements, are located within dynamic chromosomal regions.* Genes Dev. **2019**. doi:10.1101/gad.331892.119
15. Li H, Feng X, et al.. *The design and construction of reference pangenome graphs with minigraph.* Genome biology **2020**. doi:10.1186/s13059-020-02168-z
16. Human ribosomal DNA complete repeating unit. GenBank KY962518.1, 44,838 bp. NCBI Nucleotide **2018**. KY962518.1
17. Hori Y, Shimamoto A, et al.. *The human ribosomal DNA array is composed of highly homogenized tandem clusters.* Genome Research **2021**. doi:10.1101/gr.275838.121
18. Agrawal S, Ganley ARD. *The conservation landscape of the human ribosomal RNA gene repeats.* PLoS ONE **2018**. doi:10.1371/journal.pone.0207531
19. Stults DM, Killen MW, et al.. *Genomic architecture and inheritance of human ribosomal RNA gene clusters.* Genome Research **2007**. doi:10.1101/gr.6858507
20. Gibbons JG, Branco AT, et al.. *Ribosomal DNA copy number is coupled with gene expression variation and mitochondrial abundance in humans.* Nature Communications **2014**. doi:10.1038/ncomms5850
21. Gibbons JG, Branco AT, et al.. *Concerted copy number variation balances ribosomal DNA dosage in human and mouse genomes.* Proceedings of the National Academy of Sciences **2015**. doi:10.1073/pnas.1416878112
22. Stults DM, Killen MW, et al.. *Human rRNA Gene Clusters Are Recombinational Hotspots in Cancer.* Cancer Research **2009**. doi:10.1158/0008-5472.can-09-2680
23. Xu B, Li H, et al.. *Ribosomal DNA copy number loss and sequence variation in cancer.* PLoS Genetics **2017**. doi:10.1371/journal.pgen.1006771
24. Wang M, Lemos B. *Ribosomal DNA copy number amplification and loss in human cancers is linked to tumor genetic context, nucleolus activity, and proliferation.* PLoS Genetics **2017**. doi:10.1371/journal.pgen.1006994
25. Parks M, Kurylo CM, et al.. *Variant ribosomal RNA alleles are conserved and exhibit tissue-specific expression.* Science Advances **2018**. doi:10.1126/sciadv.aao0665
26. Kim J, Dillthey A, et al.. *Variation in human chromosome 21 ribosomal RNA genes characterized by TAR cloning and long-read sequencing.* Nucleic Acids Research **2018**. doi:10.1093/nar/gky442
27. Rautiainen M. *Ribotin: automated assembly and phasing of rDNA morphs.* Bioinformatics **2024**. doi:10.1093/bioinformatics/btae124
28. Altemose N. *A classical revival: Human satellite DNAs enter the genomics era.* Seminars in Cell and Developmental Biology **2022**. doi:10.1016/j.semcdb.2022.04.012
29. Hoyt SJ, Storer JM, et al.. *From telomere to telomere: The transcriptional and epigenetic state of human repeat elements.* Science **2022**. doi:10.1126/science.abk3112
30. Ting DT, Lipson D, et al.. *Aberrant Overexpression of Satellite Repeats in Pancreatic and Other Epithelial Cancers.* Science **2011**. doi:10.1126/science.1200801
31. Zhu Q, Hoong N, et al.. *Heterochromatin-Encoded Satellite RNAs Induce Breast Cancer.* Molecular Cell **2018**. doi:10.1016/j.molcel.2018.04.023
32. Bersani F, Lee E, et al.. *Pericentromeric satellite repeat expansions through RNA-derived DNA intermediates in cancer.* Proceedings of the National Academy of Sciences **2015**. doi:10.1073/pnas.1518008112
33. de Lima LG, Guarracino A, et al.. *The formation and propagation of human Robertsonian chromosomes.* Nature **2025**. doi:10.1038/s41586-025-09540-8
34. González B, Navarro-Jiménez M, et al.. *Somatic Hypomethylation of Pericentromeric SST1 Repeats and Tetraploidization in Human Colorectal Cancer Cells.* Cancers **2021**. doi:10.3390/cancers13215353

35. Altemose N, Logsdon GA, et al.. *Complete genomic and epigenetic maps of human centromeres*. Science **2022**. doi:10.1126/science.abl4178
36. Talbert PB, Henikoff S. *The genetics and epigenetics of satellite centromeres*. Genome Research **2022**. doi:10.1101/gr.275351.121
37. Choo KHA, Vissel B, et al.. *Homologous alpha satellite sequences on human acrocentric chromosomes with selectivity for chromosomes 13, 14 and 21: implications for recombination between nonhomologues and Robertsonian translocations*. Nucleic Acids Research **1988**. doi:10.1093/nar/16.4.1273
38. Logsdon GA, Rozanski AN, et al.. *The variation and evolution of complete human centromeres*. Nature **2024**. doi:10.1038/s41586-024-07278-3
39. Miga KH. *Centromeric Satellite DNAs: Hidden Sequence Variation in the Human Population*. Genes **2019**. doi:10.3390/genes10050352
40. Sullivan LL, Sullivan BA. *Genomic and functional variation of human centromeres*. Experimental Cell Research **2020**. doi:10.1016/j.yexcr.2020.111896
41. Mastrorosa F, Daponte A, et al.. *Complete chromosome 21 centromere sequencing of families with Down syndrome reveals centromere size asymmetry*. bioRxiv (Cold Spring Harbor Laboratory) **2024**. doi:10.1101/2024.02.25.581464
42. Gerton JL. *A working model for the formation of Robertsonian chromosomes*. Journal of Cell Science **2024**. doi:10.1242/jcs.261912
43. Vollger MR, Dishuck PC, et al.. *Increased mutation and gene conversion within human segmental duplications*. Nature **2023**. doi:10.1038/s41586-023-05895-y
44. Grummt I. *Life on a planet of its own: regulation of RNA polymerase I transcription in the nucleolus*. Genes & Development **2003**. doi:10.1101/gad.1098503r
45. Németh A, Conesa A, et al.. *Initial Genomics of the Human Nucleolus*. PLoS Genetics **2010**. doi:10.1371/journal.pgen.1000889
46. Koningsbruggen Sv, Gierlinski M, et al.. *High-Resolution Whole-Genome Sequencing Reveals That Specific Chromatin Domains from Most Human Chromosomes Associate with Nucleoli*. Molecular Biology of the Cell **2010**. doi:10.1091/mbc.e10-06-0508
47. Németh A, Längst G. *Genome organization in and around the nucleolus*. Trends in Genetics **2011**. doi:10.1016/j.tig.2011.01.002
48. Potapova T, Unruh JR, et al.. *Superresolution microscopy reveals linkages between ribosomal DNA on heterologous chromosomes*. The Journal of Cell Biology **2019**. doi:10.1083/jcb.201810166
49. Cerqueira AV, Lemos B. *Ribosomal DNA and the Nucleolus as Keystones of Nuclear Architecture, Organization, and Function*. Trends in Genetics **2019**. doi:10.1016/j.tig.2019.07.011
50. Rubbi CP, Milner J. *Disruption of the nucleolus mediates stabilization of p53 in response to DNA damage and other stresses*. The EMBO Journal **2003**. doi:10.1093/emboj/cdg579
51. Narla A, Ebert BL. *Ribosomopathies: human disorders of ribosome dysfunction*. Blood **2010**. doi:10.1182/blood-2009-10-178129
52. Gershman A, Sauria ME, et al.. *Epigenetic patterns in a complete human genome*. Science **2022**. doi:10.1126/science.abj5089
53. Pálsson G, Hardarson MT, et al.. *Complete human recombination maps*. Nature **2025**. doi:10.1038/s41586-024-08450-5
54. Altemose N, Noor N, et al.. *A map of human PRDM9 binding provides evidence for novel behaviors of PRDM9 and other zinc-finger proteins in meiosis*. eLife **2017**. doi:10.7554/elife.28383
55. Cheng EY, Hunt PA, et al.. *Meiotic Recombination in Human Oocytes*. PLoS Genetics **2009**. doi:10.1371/journal.pgen.1000661
56. Bell AD, Mello CJ, Nemesh J, et al.. *Insights into variation in meiosis from 31,228 human sperm genomes*. Nature **2020**. doi:10.1038/s41586-020-2347-0

57. Brown P, Judis L, et al.. *Meiotic Synapsis Proceeds from a Limited Number of Subtelomeric Sites in the Human Male*. The American Journal of Human Genetics **2005**. doi:10.1086/468188
58. Cox DM, Butler MG. *The 15q11.2 BP1-BP2 microdeletion syndrome: a review*. Int. J. Mol. Sci. **2015**. doi:10.3390/ijms16024068
59. Porubsky D, Yoo D, Dishuck PC, et al.. *Population differences of chromosome 22q11.2 duplication structure predispose differentially to microdeletion and inversion*. bioRxiv **2025**. doi:10.1101/2025.07.04.662981
60. Rhie A, Nurk S, et al.. *The complete sequence of a human Y chromosome*. Nature **2023**. doi:10.1038/s41586-023-06457-y
61. Hallast P, Ebert P, et al.. *Assembly of 43 human Y chromosomes reveals extensive complexity and variation*. Nature **2023**. doi:10.1038/s41586-023-06425-6
62. Yoo D, Rhie A, et al.. *Complete sequencing of ape genomes*. Nature **2025**. doi:10.1038/s41586-025-08816-3
63. Nielsen J, Wohler M. *Chromosome abnormalities found among 34910 newborn children: results from a 13-year incidence study in yrhush, Denmark*. Human Genetics **1991**. doi:10.1007/bf01213097
64. Page SL, Shatter LG. *Nonhomologous Robertsonian translocations form predominantly during female meiosis*. Nature Genetics **1997**. doi:10.1038/ng0397-231
65. Cassidy SB, Dykens EM, et al.. *Prader-Willi and Angelman syndromes: Sister imprinted disorders*. American Journal of Medical Genetics **2000**. doi:10.1002/1096-8628(200022)97:2<136::aid-ajmg5>3.0.co;2-v
66. Shaffer LG, Lupski JR. *Molecular mechanisms for constitutional chromosomal rearrangements in humans*. Annual Review of Genetics **2000**. doi:10.1146/annurev.genet.34.1.297
67. Salim D, Gerton JL. *Ribosomal DNA instability and genome adaptability*. Chromosome Research **2019**. doi:10.1007/s10577-018-9599-7
68. Aganezov S, Yan SM, et al.. *A complete reference genome improves analysis of human genetic variation*. Science **2022**. doi:10.1126/science.abl3533
69. Wang T, Antonacci-Fulton L, et al.. *The Human Pangenome Project: a global resource to map genomic diversity*. Nature **2022**. doi:10.1038/s41586-022-04601-8
70. Garrison E, Guarracino A, et al.. *Building pangenome graphs*. Nature Methods **2024**. doi:10.1038/s41592-024-02430-3
71. Hickey G, Monlong J, et al.. *Pangenome graph construction from genome alignments with Minigraph-Cactus*. Nature Biotechnology **2023**. doi:10.1038/s41587-023-01793-w
72. Sirén J, Monlong J, et al.. *Pangenomics enables genotyping of known structural variants in 5202 diverse genomes*. Science **2021**. doi:10.1126/science.abg8871
73. Ebler J, Ebert P, et al.. *Pangenome-based genome inference allows efficient and accurate genotyping across a wide spectrum of variant classes*. Nature Genetics **2022**. doi:10.1038/s41588-022-01043-w
74. Antipov D, Rautiainen M, et al.. *Verkko2 integrates proximity-ligation data with long-read De Bruijn graphs for efficient telomere-to-telomere genome assembly, phasing, and scaffolding*. Genome Research **2025**. doi:10.1101/gr.280383.124
75. Sanders AD, Meiers S, et al.. *Single-cell analysis of structural variations and complex rearrangements with tri-channel processing*. Nature Biotechnology **2019**. doi:10.1038/s41587-019-0366-x
76. Porubsky D, Höps W, et al.. *Recurrent inversion polymorphisms in humans associate with genetic instability and genomic disorders*. Cell **2022**. doi:10.1016/j.cell.2022.04.017
77. Gilpatrick T, Lee I, et al.. *Targeted nanopore sequencing with Cas9-guided adapter ligation*. Nature Biotechnology **2020**. doi:10.1038/s41587-020-0407-5
78. Stergachis AB, Debo B, et al.. *Single-molecule regulatory architectures captured by chromatin fiber sequencing*. Science **2020**. doi:10.1126/science.aaz1646
79. Miga KH, Koren S, et al.. *Telomere-to-telomere assembly of a complete human X chromosome*. Nature **2020**. doi:10.1038/s41586-020-2547-7

80. Jarvis ED, Formenti G, et al.. *Semi-automated assembly of high-quality diploid human reference genomes*. Nature **2022**. doi:10.1038/s41586-022-05325-5
81. Logsdon GA, Vollger MR, et al.. *Long-read human genome sequencing and its applications*. Nature Reviews Genetics **2020**. doi:10.1038/s41576-020-0236-x
82. Porubsky D, Vollger MR, et al.. *Gaps and complex structurally variant loci in phased genome assemblies*. Genome Research **2023**. doi:10.1101/gr.277334.122
83. Rautiainen M, Nurk S, et al.. *Telomere-to-telomere assembly of diploid chromosomes with Verkko*. Nature Biotechnology **2023**. doi:10.1038/s41587-023-01662-6
84. Cheng H, Concepcion GT, et al.. *Haplotype-resolved de novo assembly using phased assembly graphs with hifiasm*. Nature Methods **2021**. doi:10.1038/s41592-020-01056-5
85. Cheng H, Asri M, et al.. *Scalable telomere-to-telomere assembly for diploid and polyploid genomes with double graph*. Nature Methods **2024**. doi:10.1038/s41592-024-02269-8
86. Mastoras M, Asri M, et al.. *Highly accurate assembly polishing with DeepPolisher*. Genome Research **2025**. doi:10.1101/gr.280149.124
87. Li H. *Minimap2: pairwise alignment for nucleotide sequences*. Bioinformatics **2018**. doi:10.1093/bioinformatics/bty191
88. Rhie A, Walenz BP, et al.. *Mercury: reference-free quality, completeness, and phasing assessment for genome assemblies*. Genome biology **2020**. doi:10.1186/s13059-020-02134-9
89. Sweeten AP, Schatz MC, et al.. *ModDotPlot—rapid and interactive visualization of tandem repeats*. Bioinformatics **2024**. doi:10.1093/bioinformatics/btae493
90. Porubsky D, Guitart X, et al.. *SVbyEye: a visual tool to characterize structural variation among whole-genome assemblies*. Bioinformatics **2025**. doi:10.1093/bioinformatics/btaf332
91. Cartney AMM, Shafin K, et al.. *Chasing perfection: validation and polishing strategies for telomere-to-telomere genome assemblies*. Nature Methods **2022**. doi:10.1038/s41592-022-01440-3
92. Guarracino A, Heumos S, et al.. *ODGI: understanding pangenome graphs*. Bioinformatics **2022**. doi:10.1093/bioinformatics/btac308
93. Benson G. *Tandem repeats finder: a program to analyze DNA sequences*. Nucleic Acids Research **1999**. doi:10.1093/nar/27.2.573
94. Morgulis A, Gertz EM, et al.. *WindowMasker: window-based masker for sequenced genomes*. Bioinformatics **2005**. doi:10.1093/bioinformatics/bti774
95. Numanagic I, Gökkaya AS, et al.. *Fast characterization of segmental duplications in genome assemblies*. Bioinformatics **2018**. doi:10.1093/bioinformatics/bty586
96. Wei W, Gui S, et al.. *wgatools : an ultrafast toolkit for manipulating whole-genome alignments*. Bioinformatics **2025**. doi:10.1093/bioinformatics/btaf132
97. Noyes MD, Harvey WT, et al.. *Familial long-read sequencing increases yield of de novo mutations*. The American Journal of Human Genetics **2022**. doi:10.1016/j.ajhg.2022.02.014
98. Dolzhenko E, English AC, et al.. *Characterization and visualization of tandem repeats at genome scale*. Nature Biotechnology **2024**. doi:10.1038/s41587-023-02057-3
99. Sonay TB, Carvalho T, et al.. *Tandem repeat variation in human and great ape populations and its impact on gene expression divergence*. Genome Research **2015**. doi:10.1101/gr.190868.115
100. Smolka M, Paulin LF, et al.. *Detection of mosaic and population-level structural variants with Sniffles2*. Nature Biotechnology **2024**. doi:10.1038/s41587-023-02024-y
101. Rausch T, Zichner T, et al.. *DELLY: structural variant discovery by integrated paired-end and split-read analysis*. Bioinformatics **2012**. doi:10.1093/bioinformatics/bts378
102. Kolmogorov M, Billingsley KJ, et al.. *Scalable Nanopore sequencing of human genomes provides a comprehensive view of haplotype-resolved variation and methylation*. Nature Methods **2023**. doi:10.1038/s41592-023-01993-x

103. Behera S, Catreux S, et al.. *Comprehensive genome analysis and variant detection at scale using DRAGEN*. Nature Biotechnology **2024**. doi:10.1038/s41587-024-02382-1
104. Wagner J, Olson ND, et al.. *Curated variation benchmarks for challenging medically relevant autosomal genes*. Nature Biotechnology **2022**. doi:10.1038/s41587-021-01158-1
105. Olson ND, Wagner J, et al.. *Variant calling and benchmarking in an era of complete human genome sequences*. Nature Reviews Genetics **2023**. doi:10.1038/s41576-023-00590-0
106. Porubsky D, Dashnow H, et al.. *Human de novo mutation rates from a four-generation pedigree reference*. Nature **2025**. doi:10.1038/s41586-025-08922-2
107. English AC, Menon VK, et al.. *Truvari: refined structural variant comparison preserves allelic diversity*. Genome biology **2022**. doi:10.1186/s13059-022-02840-6
108. Huang N, Li H. *compleasm: a faster and more accurate reimplementation of BUSCO*. Bioinformatics **2023**. doi:10.1093/bioinformatics/btad595
109. Quinlan AR, Hall IM. *BEDTools: a flexible suite of utilities for comparing genomic features*. Bioinformatics **2010**. doi:10.1093/bioinformatics/btq033